

Visual Augmentation of Audio Semantics for Accessibility

Detailed technical report

Adam Gavriely

Bar-Ilan University, Department of Computer Science

MSc final project (in place of a thesis), submitted to Oren Glickman and the project advisor

2 October 2026

Code: <https://github.com/adamgavriely/MscFinalProject>, tag `detector-frozen-2026-10-02`

Abstract

Subtitles carry speech. The other sounds of a video – a siren behind the camera, a dog barking in the next room, glass breaking off screen – reach a deaf or hard-of-hearing viewer only if a captioner chose to write them down, and captioning practice says to leave out any sound whose source can be seen. This project builds and evaluates a system that closes the remaining gap automatically. Given a video, it detects the non-speech, non-music sounds, decides for each one whether its source is already visible, and shows a generated picture of the sound beside the video only while an *off-screen* sound is heard. No model is trained: the system is a chain of open foundation models (two sound detectors, two audio-language “listeners”, a third audio model used as a witness, a 27-billion-parameter vision-language model for the on-screen decision, and a text-to-image model), joined by rules that were each fixed on a development set before being tested.

The system is scored per sound against human labels on 159 video clips (71 development, 88 test). A *hit* is a picture of the right sound family that starts between 0.5 s before and 1 s after the sound; a *wrong* picture is one whose source is on screen, that belongs to another sound, or that has no sound at all; the cost per clip is $(4 \cdot \text{missed} + 2 \cdot \text{wrong})/\text{clips}$. The frozen system (D', git tag `detector-frozen-2026-10-02`) gets 29 of 58 needed sounds with 15 wrong pictures on the development set (cost 2.056) and 24 of 65 with 24 wrong on the test set (cost 2.409). The pipeline used for the September test table (the “old pipeline”) scored 18 / 51 / 3.690 and 21 / 40 / 2.909 on the same two sets. On the test set D' is cheaper than the last shipped base B (2.545) with a one-sided bootstrap $p = 0.034$ (about 0.07 two-sided). The test set was read after every shipped change, so it is a report on a seen set, not a blind confirmation; Section 9 lists this and the other threats to validity in one place.

The report gives the full development record: about 66 pre-registered rounds and well over a hundred full-pipeline variants, what each one tried and why it was rejected. The main finding is where the remaining errors sit. With every stage made perfect on the same detections the pipeline would still miss about a fifth of the needed sounds, because no model hears them; and the largest single loss is the on-screen decision, where today's open vision-language models confuse “the object is in the frame” with “the object is making this sound now”. A user study with deaf viewers was not run, so the claim that the pictures help is untested.

Contents

1	Introduction	4
1.1	Motivation	4
1.2	Problem statement	4
1.3	Research questions	4
1.4	Contributions	5
1.5	How to read this report	5
2	Background and related work	5

3	Task definition and metric	7
3.1	What counts as a needed sound	7
3.2	Matching pictures to sounds	7
3.3	Viewer cost	8
3.4	Statistics	9
3.5	Pass rules on the development set	9
4	Benchmark and labels	9
4.1	History of the data	9
4.2	The merged sets in numbers	10
4.3	Label quality	11
4.4	Held-out AudioSet-Strong sets	11
4.5	External check: DCASE 2025 Task 3	12
5	The system (D')	12
5.1	Overview	12
5.2	Stages at a glance	12
5.3	Stage 4: detection	12
5.3.1	Models	13
5.3.2	Steps in execution order	13
5.3.3	The scene check	15
5.4	What may be drawn	15
5.5	Stage 5: the on-screen gate	16
5.6	Stage 5b and 6: what the picture shows	16
5.7	Stage 6b: display	17
5.8	Scored harness versus live pipeline	17
6	Development process and history	17
6.1	Working rules	18
6.2	Phases	18
6.3	The shipped chain	19
6.4	What was tried and rejected	21
6.5	Pictures	22
6.6	Evaluation history: the judge era	23
7	Results	23
7.1	Headline	23
7.2	Statistical comparisons	24
7.3	Sensitivity	24
7.4	Where the misses and wrong pictures come from	25
7.5	How far could this pipeline go? (ceiling analysis)	25
7.6	The value of the on-screen gate (September test table)	26
7.7	Other measurements	26
8	Error analysis	27
8.1	Failure modes	27
8.2	The central limit: “present” is not “making the sound”	27
8.3	The detector floor	28
8.4	Is the gold too strict?	28
8.5	A worked example	28
9	Disclosures and threats to validity	28

10 Limitations and future work	30
10.1 Limitations	30
10.2 Future work	30
11 Reproduction	31
11.1 Code state	31
11.2 Environment and models	31
11.3 Running one video	31
11.4 Running a folder of clips on the cluster	31
11.5 Reproducing the scores	32
11.6 The Decision Inspector	32
11.7 Where the records are	32
12 Conclusion	32
A Prompts	33
B Configuration of D'	34
C Round-by-round record	35
D Glossary of internal names	37

1 Introduction

1.1 Motivation

Deaf and hard-of-hearing (DHH) viewers receive speech through subtitles and, where available, sign language. Neither channel is built for the rest of the soundtrack. Subtitles for the deaf and hard of hearing (SDH) may name a sound effect in brackets, but the DCMF Captioning Key [12] tells captioners that the source of a sound “may be omitted if it can be clearly seen onscreen”, and that background effects are written only when they matter to the plot. A study of more than 715,000 YouTube videos found hand-written non-speech captions in only 5–7 % of them, and environmental sounds in 2–3 % [31]. Field studies of sound-awareness tools [25, 23, 43] show that DHH users want to know about exactly the sounds that carry information – knocks, alarms, sirens, a baby crying, a dog barking – and interviews on non-speech captions report that viewers want the type and the timing of story-relevant sounds [32].

The information a hearing viewer gets from an ambient sound is therefore lost in proportion to how far its source is from the frame. A siren over a visible ambulance tells the viewer nothing new; a siren over a crowd looking up *is* the scene. This project targets the second case. It differs from audio-to-image generation [44, 6], which turns a whole soundtrack into a picture regardless of the video, and from sound-source localisation [42, 47], which asks where in the frame a sound comes from but not whether the viewer needs to be told about it.

1.2 Problem statement

Given a video, produce the same video with a side panel that shows, while an ambient sound is heard whose source the viewer cannot see, a picture of that sound – and shows nothing otherwise. Speech and music are never drawn (they have their own channels). Human non-word sounds such as laughter, coughing or a scream *are* drawn. No model is trained or fine-tuned: the system is an inference pipeline over open models, and human work is limited to the labels used for evaluation. The system must run on one video at a time (“online”): every rule has to work without seeing other clips, and every expensive step needs a fixed trigger.

Concretely, for every sound in a clip the system must answer four questions:

1. **Is there a sound?** Find each non-speech, non-music sound and when it starts.
2. **What is it?** Name it at a level a picture can show (for example *dog bark*, not *animal*).
3. **Can the viewer already see it?** If the thing making the sound is visibly making it, stay silent.
4. **What should the picture show, and for how long?**

1.3 Research questions

The research questions are those of the project proposal (6 August 2026); Section 12 gives the answer to each.

- RQ1. Which types of audio information contribute most to scene understanding when presented visually?
- RQ2. What level of semantic granularity should be used to represent ambient sounds?
- RQ3. Can automatically generated visual augmentations improve accessibility beyond subtitles?
- RQ4. How should semantic audio augmentation systems be evaluated?
- RQ5. How does the proposed pipeline compare with representative audio-to-visual approaches?

1.4 Contributions

1. **A task definition** with a precise rule for when a picture is due: a sound is *needed* when it is neither visible (“seen making the sound now”) nor obvious (“with the sound off, the viewer would still know it is happening”), and it is an event, not steady background (Section 3).
2. **A per-sound benchmark**: 189 human-labelled clips, of which 159 form the merged development (71) and test (88) sets used for the final numbers, with timed labels, visibility and importance per sound (Section 4).
3. **A metric that matches the decision**: hits, wrong pictures split by cause, and a viewer cost that prices a missed sound at twice a wrong picture, with a sensitivity scale for on-screen pictures (Section 3.3).
4. **A working pipeline** (D′) of open models with no training, runnable on one video (`main.py`; a ComfyUI wrapper lives in `comfyui_nodes/`) (Section 5).
5. **A complete, pre-registered development record**: every idea, its pass bar written before the run, its result and the reason it was kept or dropped (Section 6, Appendix C).
6. **An error analysis** that locates the limit of today’s open models for this task: sounds no model hears, and the confusion between an object being present and the object making the sound (Section 8).
7. **Negative results about automatic evaluation**: references written by models are circular or decide visibility at chance, and LLM judges cannot tell a picture from a text tag (Section 6.6).

1.5 How to read this report

This is a technical report in the sense of the course guidelines: it aims at a depth that lets a reader reproduce the system and its numbers. Section 2 places the work. Sections 3 and 4 define the task, the metric and the data. Section 5 describes the frozen system stage by stage with every threshold. Section 6 gives the development process and history, Section 7 the results (raw counts and derived scores), Section 8 the error analysis, Section 9 the disclosures, Section 10 limitations and future work, and Section 11 exact reproduction steps. The appendices hold the prompts, the full configuration, the round-by-round table and a glossary.

Plain-language summary. The system listens to a video and looks at it. When it hears something (for example a dog barking) it asks: can the viewer already see the dog barking? If yes, it does nothing. If no, it draws a picture of a barking dog and shows it next to the video while the bark is heard. On new test videos it finds about 37% of the sounds a deaf viewer would need, and about half of the pictures it shows are right. Most of what it misses, it misses because no current open model hears the sound under speech and music, or because the vision model believes a visible object is the source when it is not.

Repository map. The code is in `src/` (one package per stage), the entry point is `main.py`, the configuration is `config.py` (`use_shipped()` sets the frozen system), the benchmark and scoring harness are in `benchmark/` and `benchmark/gold/`, cluster job scripts in `slurm/`, and every pre-registration, panel and decision log in `docs/`. Appendix D explains the internal names (B0r, SHIP8, WW5, ...) that appear in the logs.

2 Background and related work

Accessibility of non-speech sound. The DCMP Captioning Key [12] is the practice the visibility rule of this project follows: describe a sound only when the picture does not already tell the viewer. May et al. [31] measured how rarely this happens on YouTube, and a later survey of 168

DHH viewers with interviews [32] found that they want the type, the timing and the story-relevant sounds, and want a choice. An EEG study [39] found that standard sound captions lose emotional information. Beyond Subtitles [1] interviewed 11 DHH viewers about captions and graphics for non-speech sounds and built an authoring tool on sound event detection; it supports showing *important* sounds only. Caption Royale [13] (39 DHH participants) and prosody captions [37] show that visual styling is welcome when it is readable and does not distract. Sound-awareness systems – head-mounted displays [24], home displays [25, 23], SoundWatch [43], ProtoSound [38] – show that DHH users want urgent sounds fast and are worried by false alarms; this is why a wrong picture has a cost in our metric. Earlier work found icons learnable but slower than text at first [49] and peripheral sound visualisations useful [30]. None of this work generates imagery for ambient sound in recorded video, selected automatically and gated by what is already visible.

Sound event detection and tagging. AudioSet [18] (about two million 10-s clips, an ontology of 632 classes of which 527 are released) is the label space of almost every tagger. PANNs [28], AST [20] and BEATs [9] are clip-level taggers; AudioSet-Strong [22] adds labels with 0.1-s timing. Text-queried, open-vocabulary detectors score a free-text name per frame: CLAP [15], FLAM [51], FlexSED [21] and the audio part of PE-AV [48]. Sound event detection is usually scored with event-based precision, recall and F1 with a time collar [33] or with threshold-free PSDS [5, 16]. Our metric borrows the onset-only collar but makes it asymmetric (a picture may start a little early but not late), and reports a cost rather than F1, in the spirit of cost curves [14]. A known weakness of taggers matters here: a quiet sound under speech or music is ranked below *Speech* or *Music*, and most of the sounds in our benchmark are exactly of this kind.

Audio-language models. Qwen-Audio and Qwen2-Audio [10, 11], SALMONN [45] and Qwen3-Omni [53] answer free questions about audio and are benchmarked on MMAU [40]. In this project two of them (Qwen3-Omni and Audio Flamingo Next) are used as *listeners*: they are asked whether a candidate sound is present and to list what they hear, and their answers decide whether a weak detection is kept. Their free text never reaches the picture prompt, because asked “what is making this sound?” the listener named a different maker on 35 of 54 sounds (a hen’s cluck became “goats bleating”; docs/freeze_picture_setup_2026-09-25.md).

Vision-language models. Qwen2.5-VL [3], LLaVA [29] and their successors answer open questions about frames and are the only component that can decide whether a sound’s source is visible without a fixed object list. Known failure modes matter for this task: a prior on answer letters in multiple choice [58], weak handling of negation [2], and a tendency to treat a present object as the sounding one, which recent audio-visual benchmarks also report (the project’s VLM panel notes list them, docs/review/vlm_panel_2026-10-01/f2_prompts.md). Section 8 shows that this last failure is the main limit of the gate.

Sound-source localisation and off-screen sound. Self-supervised localisation [41, 8, 42] and object-aware localisation [47] point to where in the frame a sound comes from. Juanola et al. [26] show that many of these models barely change their output for silence, noise or off-screen audio, which is the case that matters here; training with such negatives helps [27]. Off-screen separation [34, 46, 56] outputs audio, not a decision about what the viewer needs. These models answer “where?”, not “whether?”, and none was found that decides “visible and making the sound now”; two such signals (an audio-visual localiser and an audio-visual synchrony model) were tested as gate votes and rejected (Section 6).

Audio-to-image generation. Sound2Scene [44], AudioToken [54], SonicDiffusion [6], Seeing Sound [35] and AudioCanvas [17] draw an image from audio; TempoTokens [55] adapts a text-to-video model to audio. They are the natural comparison family but a poor fit: they condition on the whole clip (and so draw the scene, not the one unseen source), cannot be told to leave an object

out, and never look at the video. This project therefore uses text-to-image models (SDXL [36], FLUX.1 [4], finally Qwen-Image [52]) with a short guarded prompt, and isolates the value of the on-screen decision with a *blind* baseline that uses the same detector and generator but draws every sound.

Evaluation with model judges. LLM-as-a-judge has documented position, length and self-preference biases [57]; audio-caption metrics validated against people (FENSE [59], CLAIR-A [50]) exist for text, not for pictures shown beside a video. The proposal planned an automatic protocol (a VLM describes the augmented video, an LLM scores it against a reference). Section 6.6 reports why it was replaced by per-sound scoring against human labels.

3 Task definition and metric

3.1 What counts as a needed sound

The unit of evaluation is one sound in one clip. For every sound the annotator records a label (mapped to the AudioSet ontology), its onset and offset (about 0.5-s precision), and four judgements (`benchmark/gold/tool_template.html`, `docs/metric_per_sound.md`):

visible

the thing making the sound is *seen making it now*. A barking dog on screen is visible; a baby held quietly is not visibly crying; an alarm box on a wall is not visibly ringing. If the source is visible for only part of the sound, tick visible only if it is visible for most of it.

obvious

“with the sound off, would a viewer still know this sound is happening now?” (for example waves when the sea fills the frame).

importance

a property of the sound, rated as if the screen were black: 1 = steady noise of the place, no clear start or end; 2 = an event that can be said in one sentence; 3 = danger or a key story moment (siren, alarm, explosion, gunshot, glass, baby crying, doorbell, phone, horn, scream). When unsure, the lower value.

masked

the sound is covered by speech or music (recorded, not used in scoring).

A sound is **needed** if it is neither visible nor obvious (`benchmark/gold/score_per_sound.py`, `load_gold`), and it is **scored** if its importance is 2 or 3 (`MIN_IMPORTANCE = 2`). A needed sound of importance 1 is “don’t care”: a picture on it is neither a hit nor wrong. Speech and music are never sounds. Every heard sound is labelled, not only the needed ones, because a picture with no labelled sound under it counts as wrong.

Example. In a clip of a street at night, a police siren is heard while the camera shows a crowd looking up. The siren is not visible and not obvious from the picture, and it is importance 3: it is *needed*. If the police car then drives into view with its lights on, a second siren row is labelled visible.

3.2 Matching pictures to sounds

Pictures are scored as they are *displayed*: the scorer replays the compositor’s display rules (repeat merge, minimum time on screen, grouping, at most three slots) before matching (`load_pictures`). The rules, from `score_clip`:

1. **Onset window.** A picture can match a sound only if the picture *starts* within $[t_{\text{onset}} - 0.5\text{ s}, t_{\text{onset}} + 1.0\text{ s}]$. The end of the picture is not scored. (A scoring panel first proposed +2 s; the author set +1.0 s on 19 Sept, before any per-sound score.)
2. **Same family.** The picture’s label and the sound’s label must be the same, or one must be an ontology ancestor of the other, and both must be below the top level of the ontology (depth ≥ 1): *Vehicle* matches *Car*, but the top-level *Sounds of things* matches nothing.
3. **One to one.** Each sound takes at most one picture (the nearest in time). Extra pictures on a sound already taken are *duplicates*: not credited, not counted as wrong.

Each scored needed sound is a **hit** or a **miss**. Each picture that is not a hit or a duplicate is **wrong**, of one of three kinds:

on screen (“visible”) it matches a sound that is not needed (visible or obvious). The viewer could see it; the picture is redundant.

other sound (“cross”) no same-family sound starts in its window, but some labelled sound is playing or starting there. This includes a second picture of a long sound that started earlier and is still playing – the *strict* rule.

nothing (“phantom”) no labelled sound at all is there.

“Wrong” is therefore strict: it counts unneeded pictures, not only false ones. Section 9 gives a human check of how often a “wrong” picture is in fact a real sound the labels missed.

3.3 Viewer cost

The decision metric of the final rounds is the viewer cost per clip,

$$C = \frac{4 \cdot \text{\#missed} + \beta \cdot \text{\#wrong}}{\text{\#clips}}, \quad \beta = 2, \quad (1)$$

lower is better (COST_MISS, COST_FA = 4.0, 2.0 in `score_per_sound.py`). A missed needed sound costs twice a wrong picture: one hit pays for at most two wrong pictures, so a picture is worth showing when the chance that it is right exceeds $\beta/(4 + \beta) = 1/3$. The weights come from the rubric of the September judge (a withheld picture scored 0 where one was due, a redundant one about 3 of 4) and were declared for a gate sweep before any per-sound result; they were promoted to a reported outcome only after per-sound F1 came out null (amendment 9, 22 Sept), so the choice of $\beta = 2$ is *post hoc* and is reported with sensitivity.

On-screen weight w . A picture of a sound whose source is on screen is less harmful than a picture of nothing. From 1 Oct the cost is also reported as

$$C_w = \frac{4 \cdot \text{\#missed} + w \cdot \text{\#visible} + 2 \cdot \text{\#cross} + 2 \cdot \text{\#phantom}}{\text{\#clips}}, \quad w \in \{0, 1, 2\}, \quad (2)$$

with $w = 2$ (Eq. 1) as the primary. Wins that hold only at $w = 1$ were to go to a separate “more hits” profile, never to the main system (`docs/prereg_round13_detector_push.md`, line 2050).

Why not F1. Per-sound F1 on the same labels was the pre-registered headline from 22 Sept. It weighs a miss and a wrong picture equally, and the gate’s value lies almost entirely in the wrong pictures it removes, so the F1 difference between the gated system and the blind baseline was null in every version (for example +0.059 [−0.030, +0.144], $p = 0.18$, on the September test table) while the number of wrong pictures fell significantly (Section 7.6). F1, precision and recall are still computed by the scorer.

3.4 Statistics

All comparisons are paired by clip. A **paired clip bootstrap** (2000 draws, seed 0) resamples clips and scores every system on the same resampled clips. Two read-outs are used, and the report always says which:

- *Detector rounds* (`benchmark/gold/dev_candidates_check.py`, `boot`): mean per-clip cost difference d , a 95 % percentile interval, and a **one-sided** p = share of draws with $d \geq 0$. All p -values of the 29 Sept–2 Oct detector push (0.034, 0.017, 0.015, ...) are of this kind. A two-sided value is about twice as large.
- *September test table* (`benchmark/gold/holm_table.py`): **two-sided** $p = \min(1, 2 \min(P(d \geq 0), P(d \leq 0)))$ with Holm step-down correction at $\alpha = 0.05$ over two pre-declared families (7 and 3 comparisons).

With 88 test clips the smallest cost difference this design can detect is about 0.19 per clip; a difference of 0.09 would need about 385 clips (prereg line 902). Most single changes in Section 6 are smaller than that, which is why decisions were made on pre-declared rules on the development set and not on test significance.

3.5 Pass rules on the development set

A change was adopted (“shipped”) only if it passed a rule written before its run, on the merged development set:

main rule hits $\geq$ base, wrong $\leq$ base + 2×(hits gained), cost $<$ base, and no needed hit lost on either part of the development set;

fewer-pictures clause (from 30 Sept, 16:02) cost $<$ base, at least three wrong pictures removed per hit lost, and at most three hits lost;

held-out screen rules about the audio part could be screened first on 415 held-out AudioSet-Strong clips with timed labels, never seen by the selection (Section 4.4).

From 1 Oct the standing rule was: a pre-registered development pass ships, and the test set is read once afterwards and only reported. Section 9 counts how often that happened.

4 Benchmark and labels

4.1 History of the data

The benchmark went through three stages. Table 1 lists every set used in the project.

Clip-level benchmark (August–September). 274 short clips (10–20s) were labelled by the author into four scenarios: *unseen ambient* (an ambient sound whose source is off screen; 48 clips), *mixed* (some sources visible, some not; 25), *seen ambient* (every ambient source on screen; 132) and *no ambient* (speech or music only; 69) (`benchmark/tags.json`). Clips came from Wikimedia Commons (CC), the Internet Archive (public domain), UnAV-100 [19] (CC-BY), VGGSound [7] controls and YouTube (`benchmark/README.md`). Four sourcing strategies (keyword scraping, AudioSet strong segments, curated film scenes, UnAV-100 overlap moments) and an automatic filter all landed at 15–20 % usable clips, which is why the set stopped at 274 rather than the planned 300. A test set of 100 clips (25 per scenario) was fixed. The only reliability number for these labels is intra-rater: 60 clips relabelled blind, 78 % agreement, Cohen’s $\kappa = 0.60$.

Per-sound gold (22 September). After the judge-based evaluation was abandoned (Section 6.6), the author labelled clips *per sound* with the fields of Section 3.1 (`benchmark/gold/annotations/gold_AG.json`). The first export had 139 clips: 109 benchmark clips plus 30 “slice B” clips from the AudioSet-Strong evaluation split in which a consequential sound is at least half covered by

Table 1: Data sets. “Needed” = needed sounds of importance 2–3 (the scored denominator). Counts for the gold sets were recomputed with the scorer’s own loader (`score_per_sound.py`) on the current gold file.

set	clips	needed	what it is / how it was used
clip-level benchmark	274	–	four scenarios, clip labels; v1–v3 judge era (Aug–16 Sept)
DEV-49	49	38	gold clips from the old 100-clip test set; development
TEST-60	60	43	gold clips labelled after the freeze; September test table
DEV2 / TEST2	22 / 28	20 / 22	tagger clips (31 web, 19 AudioSet-Strong)
merged DEV	71	58	selection set for every final decision
merged TEST	88	65	reported once per shipped change
slice B	30	31	AudioSet-Strong clips with a masked sound; reported separately
AudioSet calibration	280	–	fitting set for detector rounds 4–12 (retired, round 12)
held-out AudioSet	415	–	screening set for audio rules (252 complex, 163 random)
fresh confirmation	422	–	built 28 Sept for a final detector check; never scored
DCASE 2025 Task 3	258 ev.	–	external check of the September gate (on/off-screen labels)

speech or music. The 109 benchmark clips were split by provenance: **DEV-49** = the gold clips that had been in the old 100-clip test set (already touched by earlier runs), and **TEST-60** = the clips labelled after the pipeline was frozen. Slice B is always reported separately.

Tagger clips and the merged sets (29 September – 1 October). To make both halves larger, new clips were labelled with a stand-alone tool (`tagger/`): 100 YouTube clips (d001–d100, one 15–20-s cut per upload, found by scene type: transport, city walk, market, nature, home, incident, work event, water, film/TV) and 50 AudioSet-Strong evaluation clips (d101–d150, 10 s). 50 clips were labelled in three batches and split into DEV2 (22) and TEST2 (28) by a hash of the clip name (batch 1) or by balanced pairs (batches 2–3). On 30 Sept the halves were merged: **merged DEV = DEV-49 + DEV2 = 71 clips** and **merged TEST = TEST-60 + TEST2 = 88 clips**. All final numbers in this report are on these two sets.

4.2 The merged sets in numbers

Table 2: Composition of the merged sets (recomputed from `gold_AG.json`).

set	clips	sounds	needed	imp. 3	clips w. needed	unseen/mixed/seen/none
DEV-49	49	95	38	14	24	16/9/16/8
TEST-60	60	94	43	11	28	13/15/12/20
DEV2	22	59	20	3	14	8/7/7/0
TEST2	28	60	22	2	15	8/8/10/2
merged DEV	71	154	58	17	38	24/16/23/8
merged TEST	88	154	65	13	43	21/23/22/22

Merged DEV is 49 old benchmark clips, 13 web tagger clips and 9 AudioSet-Strong tagger clips; merged TEST is 60 old clips, 18 web and 10 AudioSet-Strong. Of the 109 benchmark clips, 81

have a source record (50 YouTube of which 12 are film scenes, 13 AudioSet-Strong, 7 UnAV-100, 7 Wikimedia Commons, 4 Internet Archive); 28 older clips have none. Most clips are for research use only and are not redistributed.

Needed-sound count history. The merged DEV count moved during the last week: 55 at the first merged scoring, 58 after the author’s blind visibility re-check (three sounds changed from seen to needed), 57 after a control round, and back to 58 after a correction (Section 4.3). Every system was re-scored on each version. The test labels were never changed.

Three unreachable sounds. The gold loader keeps every non-speech label, while the shipped label filter (Section 5.4) never draws three of the needed DEV labels: *Clang* in `ambient_citywalk_nyc_2627` and two *Whack*, *thwack* rows in `b3_golf_course`. They are misses for every system by construction; without them merged DEV would have 55 needed sounds.

4.3 Label quality

All gold labels are the author’s. A packet of 30 clips for a second annotator (stratified, 15 DEV + 15 TEST) was built on 27 Sept (`benchmark/gold/build_second_annotator.py`) but no labels came back, so inter-rater agreement is unknown. The tagger package was also sent to a hired annotator; no export from that person is in the gold. Three further facts bear on label quality:

- **Anchoring.** Benchmark clips were pre-filled from the system under test, and the tagger tool showed detector suggestions on 37 of 50 clips. Batch 3 (AudioSet-Strong clips) was labelled with the AudioSet labels visible behind a button on 16 of its 19 clips.
- **Visibility re-check.** On 30 Sept the author re-judged, blind to the old tick, 14 borderline DEV sounds where the pipeline and the gold disagreed (9 visible-wrong pictures, 5 gate-silenced needed sounds). Three changed from seen to needed (`as_church_bell` Bell 0.0s, `as_fire_alarm` Alarm 9.0s, `tg_d088` Thunder 1.3s). A control round on 20 random DEV sounds changed 0 of 20 (first read 1/20, then corrected by the author). The first round’s changes are therefore partly a selection effect; both rounds were applied and are disclosed (`docs/prereg_round13_detector_push.md`, lines 1841–1930).
- **Gold scope (Round 46).** The author rated, blind, pictures counted “wrong”: about 1 in 6 is a real, wanted, off-screen sound the gold does not list (2 of 11 and 3 of 20 in two samples). No gold edit was made; absolute costs are therefore slightly pessimistic for every system alike.

The tagger guide also added two rules the earlier gold tool did not state: a sound that stops for more than about 2s and comes back gets a new row, and a long sound whose visibility changes is split into two rows.

4.4 Held-out AudioSet-Strong sets

For audio-only questions (“is this candidate a real sound?”) the project used clips from the AudioSet-Strong evaluation split, whose 0.1-s labels need no new annotation. A *calibration* set of 280 clips was used in rounds 4–12 with a cost on timed labels. Round 12 found that under a corrected version of that cost the shipped detector was worse than showing nothing (1.650 vs 0.986 per clip, $d = +0.664 [+0.379, +0.950]$): AudioSet labels only a few consequential sounds per clip, so every correct but unlabelled detection counts as false. The harness was retired as a decision test (`docs/prereg_round12_v2.md`). A separate *held-out* set of 415 clips (seed 23; 252 “complex” clips where speech or music covers half the clip and a sound lies under it, 163 random) was kept for screening single rules in the October rounds, for example “how often is a BEATs detection that no listener names a real sound?” (22%; Section 8). A *fresh* set of 422 clips was built for one last confirmation and never scored.

4.5 External check: DCASE 2025 Task 3

For the September gate the DCASE 2025 Task 3 data set, which marks every event as on- or off-screen, was used as an outside check. On 258 events the Qwen2.5-VL-7B gate of that time agreed with the label 66.7% of the time: it almost never silenced an off-screen sound (97%) but silenced only 35% of on-screen ones (`benchmark/eval_dcase_visibility.json`). The DCASE label is geometric (the source direction falls inside a 100-degree field of view), not “the viewer can see it making the sound”, so it only bounds the gate loosely.

5 The system (D')

This section describes the frozen system D' exactly as the code runs it. D' is the configuration `ARMS["SHIP8+MD3+WW5+SL"]` of the scoring harness (`benchmark/gold/round13_dev.py`, line 243) and `config.use_shipped()` of the live pipeline (`config.py`, lines 503–572), frozen on 2 Oct 2026 at 01:20 UTC under git tag `detector-frozen-2026-10-02`. Where the scored harness and the live pipeline differ, the difference is stated (Section 5.8). Every value below gives its configuration variable; Appendix B lists them all.

5.1 Overview

Figure 1 shows the data flow. In words, for one video:

1. **Hear candidates.** Two sound detectors listen to the whole clip: BEATs (a 527-class tagger) and FlexSED (a detector that finds sounds by name). Anything they hear strongly enough, for at least 0.3 s, becomes a candidate with a start and an end.
2. **Check candidates.** A chain of vetoes removes candidates that only one model claims. Weak sounds that the detectors almost missed get second opinions from two audio-language “listeners” (Qwen3-Omni and Audio Flamingo Next) and a third audio model (DASM).
3. **Witness rule.** A remaining candidate needs a witness: DASM clearly hears it, or both listeners name it, or one listener names it *and* the vision model finds the sound plausible in the scene.
4. **On-screen check (the gate).** A vision-language model (Qwen3.8-27B) looks at the frames around each sound. If the thing making the sound is visibly making it, no picture is shown.
5. **Clean the display.** Repeats close together become one picture; a listener decides whether two same-type pictures up to 8 s apart are one continuing sound; a picture is dropped when its event is visibly happening.
6. **Draw.** A short, guarded text prompt is written for each remaining sound; Qwen-Image draws it; a checker asks whether the picture shows the intended thing and redraws up to five times.
7. **Compose.** The pictures are shown in up to three fixed slots beside the video, each while its sound plays (at least 1.5 s).

5.2 Stages at a glance

5.3 Stage 4: detection

Stage 4 is where most of the development effort went. It turns the audio into a list of timed candidate sounds, each with a family label and a confidence. The entry point is `detect_events` (`src/stage4_audio_event_detection/__init__.py`); unions and vetoes are in `fuse_flexsed`, the filters on rescued spans in `filter_rescued`.

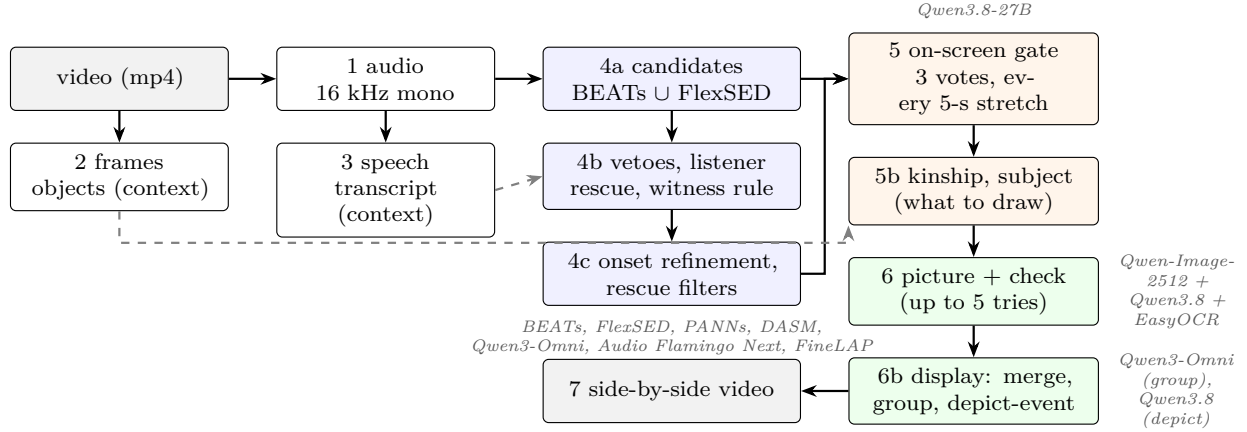


Figure 1: The D’ pipeline. Blue: detection (stage 4); orange: the on-screen decision (stage 5); green: pictures and display. Dashed: context only (objects and transcript never decide on their own). The scene check of the witness rule in stage 4b also uses Qwen3.8-27B.

5.3.1 Models

BEATs

iter3+ fine-tuned on AudioSet-2M (checkpoint `BEATs_iter3_plus_AS2M_finetuned_on_AS2M_cpt2`).

It scores a whole window, so it is run on 2-s windows every 0.25 s; the first two seconds are mirror-padded so a sound at the very start can be heard; each window’s score is stamped 0.5 s before the window ends, so a detected start is never earlier than the true one.

FlexSED

an open-vocabulary, frame-level detector [21]: a self-supervised audio encoder plus a CLAP text encoder (`laion/clap-htsat-unfused`), trained on AudioSet-Strong. It is asked one plain query per family over the 215 *depictable* families (`benchmark/gold/depictable_vocab.json`), 25 frames per second. It hears quiet sounds under speech and music far better than BEATs.

PANNs CNN14

(sound-event version, 32 kHz) used only as a clip-level veto.

DASM

Detect Any Sound Model (ACM MM 2025; weights `CPF2/detect_any_sound`), queried with “sound of ⟨family⟩” over the same 215 families, 50 frames per second. Used as a witness and a vote.

Listeners

Qwen/Qwen3-Omni-30B-A3B-Instruct and `nvidia/audio-flamingo-next-hf`. Each is asked a yes/no question (“Is the sound of ⟨family⟩ present in this recording?”, scored as the logit of *yes* minus the logit of *no*) and an open inventory (“List every distinct non-speech sound you hear in this recording, one per line, most prominent first.”). An inventory line names a family if it matches its name, an AudioSet synonym, a parent or a child, or else has sentence-embedding cosine > 0.6 (`all-mpnet-base-v2`). The audio cut is the candidate span ± 1 s, at least 1 s long.

FineLAP

(`AndreasXi/FineLAP`, MIT) a frame-level audio-text model used as a last veto on rescued sounds.

5.3.2 Steps in execution order

Table 4 gives the 16 steps. Two words are used throughout: a *rescued* span is one added by a listener after the detectors missed it; a *twinned* span is one both detectors raised.

Table 3: Stages, models and key parameters of D' (order of `src/pipeline.py`, `run`).

#	stage	model / tool	key parameters
1	audio	ffmpeg	mono WAV, 16 kHz (<code>SAMPLE_RATE</code>)
2	frames, objects	OWLv2 (<code>owlv2-base-patch16-ensemble</code>)	6 frames, threshold 0.20; context only, never silences a sound
3	speech	faster-whisper base , int8	transcript used only to ask whether people react to a sound; never drawn
4	sound events	BEATs iter3+ AS2M; FlexSED; PANNs CNN14; DASM; Qwen3-Omni-30B-A3B; Audio Flamingo Next; FineLAP	16 ordered steps, Table 4
5	on-screen gate	Qwen/Qwen3.8-27B, greedy, thinking off	6 frames per ≤ 5 -s stretch, 3 votes, majority; silenced only if every stretch is visible
5b	subject	Qwen3.8-27B + word lists	2–6 word phrase; speech text never used
6	picture	Qwen/Qwen-Image-2512	1024^2 , 50 steps, true CFG 4.0; check-and-redraw ≤ 5 tries
6b	display	Qwen3-Omni (group), Qwen3.8 (depict)	merge gap 2.5 s, min. 1.5 s on screen, ≤ 1 s past the end, group ≤ 8 s, 3 slots
7	composite	ffmpeg libx264	720-px panel, 25 fps

Table 4: Stage 4 of D', step by step (values as shipped).

#	step	rule	variable(s)
1	BEATs spans	a span of any class where the window score ≥ 0.175 ; kept if ≥ 0.3 s long; confidence = peak	AED_THRESHOLD 0.175, AED_MIN_DUR 0.3
2	FlexSED spans	same extractor on FlexSED frames at bar 0.8, ≥ 0.3 s	FLEXSED_BAR 0.8
3	twin union	a FlexSED span with a same-family BEATs span within 1 s is merged into it (earlier start kept; confidence = the stronger side, rescaled); otherwise kept as FlexSED-only	TWIN_MAX, UNION_START min
4	mirror veto + listener keep	a BEATs-only span is dropped if FlexSED's top query in its window is another family ≥ 0.7 while its own family is < 0.4 ; put back if the Qwen listener accepts it and an inventory names it	MIRROR_VETO 0.7, KEEP_NEEDS_V4
5	masked-weak veto	a BEATs-only span with confidence < 0.5 under speech or music (≥ 0.3) is dropped unless a listener accepts it	MASKED_WEAK_VETO
6	DASM clip veto	a span whose family DASM never reaches 0.0839 anywhere in the clip is dropped unless a listener keeps it	DASM_CLIP_VETO
7	witness rule	pass if DASM ≥ 0.35 for the family within the span ± 0.5 s; else pass if <i>both</i> listener inventories name the family; if exactly one does, ask the scene check (Section 5.3.3); otherwise drop	DASM_LOCAL_VETO 0.35, DASM_LOCAL_KEEP both, SCENE_FIT_LOGIT
8	inventory keep (K4A-D)	a span with confidence ≥ 0.35 for which both inventories exist and neither names its family (or a descendant) is dropped, unless DASM ≥ 0.575 within ± 0.5 s	KEEP_NEEDS_V4_ALL onto, ..._DASM_KEEP

#	step	rule	variable(s)
9	band-twin pull	the start moves back to an earlier same-family FlexSED run (≥ 0.5) that ends ≤ 1 s before and starts ≤ 1.5 s before it	BAND_TWIN_PULL 0.5
10	continuation veto	a span is dropped when a same-family FlexSED run (≥ 0.5) began ≥ 1.5 s earlier and is still on at its start (it is a later piece of a sound already shown)	CONTINUATION_VETO 0.5
11	FlexSED veto	a span whose family FlexSED never reaches 0.3 in the clip is dropped	FLEXSED_VETO 0.3
12	PANNs veto	a FlexSED-only span whose family PANNs never reaches 0.05 is dropped unless a listener accepts it	PANNS_VETO 0.05
13	listener band rescue	a FlexSED run with peak in $[0.5, 0.8)$ and no same-family span is added if accepted: peak ≥ 0.6 needs the Qwen inventory; < 0.6 needs both inventories	LISTENER_RESCUE, LISTENER_RULE TIER
14	DASM rescue	a DASM run (≥ 0.575) with no detector span nearby is added if both inventories name the family and the family is new in the clip	DASM_RESCUE, ..._NEW_ONLY
15	onset refinement	for BEATs spans: silence the first t seconds of the first firing window for 26 values of t and take the first t at which 10% of the class evidence is gone; never earlier than the detected start	ONSET_CAM, ONSET_MONOTONE
16	rescued-span filters	on rescued spans only: FineLAP ≥ 0.329 ; DASM ≥ 0.575 within ± 0.5 s; at most one rescued span per family per clip (the earliest)	FINELAP_VETO 0.329, LISTENER_DASM_VOTE, LISTENER_ONCE

The DASM thresholds 0.0839 and 0.575 were calibrated in detector round 6 (`benchmark/detector_round6.json`); the FineLAP bar 0.329 in round 31. Steps 4–8 and 12–16 depend on listener answers that are computed from a first pass with the old pipeline (the candidate “pools”), so the shipped detector cannot run without that pass (Section 11).

5.3.3 The scene check

When exactly one listener names a weak sound, the witness rule asks the vision model whether the sound is plausible in the scene. The span is cut into stretches of about 5 s; for each, six frames from 1 s before to 1 s after are shown with the question “Could the sound of $\langle \text{label} \rangle$ plausibly be heard in this scene? Answer yes or no.” and its negated twin. The answer is read from the first output token as $d = m(Q) - m(\bar{Q})$, where $m = \max$ logit over {yes, Yes} minus $\max$ logit over {no, No}; $d > 0$ counts as yes, and the sound passes if most stretches say yes. This logit readout (Round 60L) replaced a text answer that was cut off after four tokens on 10–15% of calls; it is the only difference between the system D and D’ (Section 6).

5.4 What may be drawn

The label filter (`LABEL_FILTER = "depictable", src/labels.py, is_salient_nonspeech`) maps every label to its family and removes: speech with words and its descendants (Speech, Conversation, Narration, Hubbub, Whispering, ...), music and singing and their descendants, the “channel, environment and background” branch, silence and generic labels, textures (Breathing, Rumble, Hum, Whir, Rustle, Wind and its descendants), and labels with no clear source (Sound effect, Onomatopoeia, generic impacts). Everything else is drawable, including non-word vocal events (shout, scream, laughter, cough, snoring). FlexSED and DASM are asked only about the 215 drawable families.

5.5 Stage 5: the on-screen gate

Stage 5 first groups the detections by family at the display bar 0.35 (DISPLAY_THRESHOLD); firings of one family less than 1 s apart are one burst. It then asks Qwen3.8-27B, with greedy decoding and thinking off, for every burst:

1. The burst is cut into $k = \max(1, \text{round}(\text{length}/5))$ stretches; each stretch gets six frames, evenly from 1 s before to 1 s after it.
2. Three votes per stretch (prompts in Appendix A):
 - name** “Name the thing in these frames that is making that sound ... or answer exactly: nothing.” A name that shares a word with the label counts yes; otherwise a text-only question asks whether that thing could make the sound.
 - a/b** “(a) you can SEE ⟨label⟩ happening on screen – the source is in the frames and visibly making that sound; (b) ⟨label⟩ is not visibly happening.” Asked in both option orders; if the two answers disagree the vote abstains.
 - describe** a 25-word description of the frames; yes if it names the sound or the named thing, else the same text-only question.
3. A stretch is visible if yes-votes outnumber no-votes. The sound is silenced only if *every* stretch is visible; otherwise its picture is kept for the whole burst.
4. **Kinship.** A sound is also silenced if a silenced sound of the same source overlaps it in time (shipped: only when the visible label is the same or more specific; Section 5.8).
5. **Speech context.** If at least four words of speech fall within 3 s of a sound, a text-only question asks whether the people react to it; a yes gives the sound priority for a slot and can rescue a sound just below the display bar. Speech never enters a picture.
6. **Disambiguation.** Two different labels with starts and ends within 0.6 s are one sound with two names; the frames are shown with “which is it, (a) A or (b) B or (c) cannot tell?” in both orders.

The rule “silence only if every stretch is visible” leans toward showing: a sound whose source leaves the frame for one stretch keeps its picture.

5.6 Stage 5b and 6: what the picture shows

Subject. The picture prompt is built from the detector’s label, not from free text. A resolve step asks the VLM for at most two qualifier words plus the head word of the source (accepted only if it ends with the head word, names no person and no forbidden kind), then a 3–6-word event phrase. Word-list guards remove persons, places and makers from other ontology branches. Action labels are drawn as their maker (*bark* → a barking dog) from a hand table, ontology senses (*Honk* = goose), or a closed VLM choice on the frames. Sounds that are best shown as a word (*Whoosh*, *Thud*, *Bang*, *Smack*, *Whack*) get a fixed comic burst card, and a few sounds a fixed template (thunder = “one large lightning bolt striking down from a dark storm cloud”).

Generation. The prompt is the subject plus a fixed tail (“the whole thing fully in frame, caught at the moment it makes the sound with its visible effect, one large subject filling the picture, plain white background”) and a negative prompt (scenery, landscape, room interior, street, buildings, text, letters, watermark). Qwen-Image-2512 draws it at 1024×1024 with 50 steps and true CFG 4.0; the seed is a hash of clip, label and start time, so a run is repeatable.

Check and redraw. The same Qwen3.8-27B is shown the picture with a shuffled multiple-choice question (the intended thing, known look-alikes, generic confusions such as “a person” or “a building”, “something else”); EasyOCR rejects a picture that contains a readable word. A failed

picture is redrawn with a new seed and the confusing thing added to the negative prompt, up to five tries; if all fail, a word card with the sound’s name is shown. The check was validated on known cases (7 of 7 known-bad pictures caught, 33 of 33 deliberately wrong ones refused, 3 of 68 good ones refused, 113 of 115 answers unchanged on reshuffling) but never calibrated against a human rater (Section 9).

5.7 Stage 6b: display

1. **Depict-event** (Round 57): a picture is dropped if, on six frames around its start, the VLM says its event is visibly happening (and its negated twin agrees) *and* a maker the gate named could be mistaken for the sound.
2. **Repeat merge**: bursts of one label join into one appearance if the gap from the previous real end is ≤ 2.5 s (`MERGE_GAP`).
3. **Dwell**: each appearance stays at least 1.5 s (`MIN_DWELL`) and at most 1 s past the real end of the sound (`MAX_AFTER_END`).
4. **Grouping** (Round 47): for two same-family pictures whose gap is at most 8 s, Qwen3-Omni hears the audio from just before the first ends to just after the second starts and is asked whether the second is “the same continuing sound ... (one alarm or engine with a short pause)” or “a new, separate event (... a second bark, knock or shot)”; both option orders must say “same” to merge.
5. **Slots**: at most three pictures at once, each in a fixed slot; if more overlap, sounds people react to come first, then confidence. A slot is empty when its sound is silent.

5.8 Scored harness versus live pipeline

The numbers in this report come from the DEV/TEST harness, which runs the pipeline’s own stage-4 functions on cached model scores and replays the gate answers of a full run. A parity check (`benchmark/gold/parity_check.py`) and an independent re-export of every decision (the Decision Inspector, Section 11) reproduce the published counts exactly (0 of 159 clips differ). The live configuration `use_shipped()` differs from the scored one in four known ways:

- **Kinship direction**. Live: a visible label silences only the same or a more specific label. Scored: either direction. This was a deliberate logic fix made after the scored tables (`docs/prereg_v4.md`, line 2240); its effect on D' was not re-measured.
- **Picture end**. Live pictures end at most 1 s after their sound; scored pictures had no cap. Starts – the scored quantity – are identical; on 3 TEST clips a picture ends 0.04–0.38 s earlier live.
- **Subject path**. The scored runs used the older depiction path, which can drop a sound whose depiction fails twice and feeds a similarity-based merge; the live path never drops a sound at this step. No end-to-end equality test of `main.py` against the scored numbers exists.
- **Pictures**. Scored runs used placeholder or FLUX.1-schnell pictures; the live system uses Qwen-Image with the check. Pictures are not part of the per-sound score.

6 Development process and history

The system was not designed in one step. It reached D' through fourteen phases and, in the last week, more than sixty numbered, pre-registered rounds. This section explains the working rules, then the history in three tables: the phases (Table 5), the chain of shipped changes (Table 6) and the main rejected ideas (Table 7). Appendix C has one row per round.

6.1 Working rules

From 13 Sept the following rules governed every change (`docs/plan_robustness.md`, `docs/prereg_round13_detector_push.md`):

1. **Pre-register.** Before a run, write the idea, the exact rule, the data it is decided on and the pass bar, and commit it. Results are appended below the pre-registration, never edited into it.
2. **Decide on development data only.** The development set (or, for audio-only rules, the held-out 415) decides. The test set is read after a change ships and is only reported. The author had to restate this rule on 1 Oct after a test-informed choice (Section 9).
3. **General rules only.** A rule must make sense for any video, run on one video at a time, and not name a clip or a single sound class (a rule that only ever fires on bells was rejected for this reason, Round 64).
4. **Compare with the shipped pipeline.** Every candidate is scored as a full-pipeline variant on the same clips as the current shipped system, not on a proxy; a component screen may stop an idea early but cannot ship it.
5. **Hits first.** Losing a few hits is acceptable only if many wrong pictures go (the fewer-pictures clause, Section 3.5).
6. **Consult and record.** Major choices were reviewed by panels of independent model reviewers (Claude instances with assigned roles, two to five per panel, one to four rounds), whose reports are in `docs/review/` and `docs/panel_*.md`; every decision and its reason is in a dated log.

6.2 Phases

Table 5: Project phases. Scores in each row are in that phase’s own metric and are not comparable across rows.

dates	system	scored by	outcome and why it changed
06 Aug	proposal: 7 stages (audio, VLM, ASR, BEATs/PANNs, reasoning, FLUX/SDXL, evaluation)	planned: VLM describes, LLM writes reference, LLM judges	pivot from an earlier speech-illustration project
07 Aug	v1: PANNs (bar 0.15) → Openverse image retrieval → side panel; no gate	demo clips	needed a “not seen” decision
07 Aug–07 Sep	CLIP zero-shot visibility gate; clip-level benchmark (274 clips)	clip-level gate accuracy	CLIP gate 49.3%: 71 of 103 seen clips still augmented – “object presence is not event visibility”
08–11 Sep	PANNs + OWLv2 + per-sound VLM gate (Qwen2.5-VL-7B, “name it, don’t confirm”); SDXL	100 clips × 3 systems, Mistral-7B judge 0–4, model-written reference	blind 2.94 > caption 2.68 > gated 2.41; retrieval beat SDXL by +0.17/+0.18; reference later found circular

dates	system	scored by	outcome and why it changed
12–17 Sep	v3: BEATs (replaced PANNs after 11 of 14 complaints traced to its labels), hysteresis + occlusion onsets, 7B gate with 3 votes, FLUX.1-schnell	human-grounded judge reference, dev 174 / test 100	tie 3.17 vs 3.16; gate wins where nothing is due (+0.26), loses where a picture is due (−0.24)
18–21 Sep	v4: one state-of-the-art swap per stage; Qwen3.8-27B gate; FLAM and PretrainedSED detectors tried; rubric-enforced judge	judge with a cap on redundant pictures	PretrainedSED rejected three times; 13 silent bugs found (amendment 3), all earlier judge rows kept as history only
22–23 Sep	per-sound gold (139 clips); BEATs $\cup$ FlexSED (bar 0.8), FlexSED veto 0.3, PANNs veto 0.05	per-sound F1, then viewer cost (amendment 9)	F1 difference null (+0.028); wrong pictures and precision significant
24–27 Sep	monotone onsets, no length cap; final September test table (amendment 21); Qwen-Image frozen as picture setup	Holm-corrected table on TEST-60	“Holm wins, F1 null” (Section 7.6)
27 Sep	detector rounds 1–3: cascade, PE-A-Frame, Qwen3-Omni as verifier	DEV-49 + AudioSet 280 / 415	none passed; detection declared closed, reopened next day
28 Sep	rounds 4–12 on AudioSet sets: EAT, Dasheng, DASM, SAM-Audio, 17 ideas, local contrast, DASM rescue, masker tests	cost on AudioSet labels	all failed on the 415; round 12 showed the AudioSet cost cannot track the task: harness retired
28–30 Sep	rounds 13–14: listener rescue (Qwen3-Omni, Audio Flamingo Next) + DASM vote; tagger clips added	DEV-49, then merged DEV	first gains in hits (TO1+F7F8)
30 Sep–01 Oct	ship chain SHIP1–SHIP8 on merged DEV/TEST	merged DEV, pass rules	DEV 18/51/3.690 $\rightarrow$ 28/21/2.282 (Table 6)
01–02 Oct	rounds 46–66: display rules, witness rule, scene check, many gate ideas	merged DEV, held-out 415	D’: DEV 29/15/2.056; frozen 2 Oct 01:20 UTC

6.3 The shipped chain

Table 6 is the backbone of the history: every change that entered D’, in order, with its development and test numbers on the merged sets. Development numbers use the corrected gold

(58 needed sounds); test numbers are the reads made after each change shipped. Sources: docs/review/improvements_log_2026-09-30.md, docs/review/detector_final_2026-10-02.md, docs/prereg_round13_detector_push.md.

Table 6: Changes that shipped, from the old pipeline to D'. Hits / wrong / cost; DEV hits out of 58, TEST out of 65. The p in the TEST column is one-sided, against the old pipeline B0r for SHIP5–SHIP8 and against B for D and D'.

step (name in logs)	what it adds	merged DEV	merged TEST
old pipeline (B0r)	September test-table system: BEATs $\cup$ FlexSED, FlexSED and PANNs vetoes, Qwen3.8 gate	18 / 51 / 3.690	21 / 40 / 2.909
TO1+F7F8 (R14)	listener rescue of weak FlexSED runs (tiered: Qwen alone above 0.6, both listeners below), one rescue per family, mirror veto with listener keep, DASM vote on rescues	26 / 45 / 3.070	22 / 38 / 2.818
+ N2b, DR2, K-V4 (R16–21)	masked-weak veto; DASM rescue of new families; a listener keep must also be named in an inventory	27 / 38 / 2.817	23 / 36 / 2.727
+ DV (R28)	drop a sound DASM never hears in the clip	27 / 31 / 2.620	23 / 34 / 2.682
+ BTP (R30)	start a late picture at an earlier weak run of the same sound	28 / 30 / 2.535	23 / 33 / 2.659 (p 0.033)
+ CONT (R31)	drop a picture that is only a later piece of a sound already going on	28 / 25 / 2.394	24 / 32 / 2.591 (p 0.025)
+ FineLAP veto (R31)	a 2025 frame-level model double-checks rescued sounds	28 / 24 / 2.366	24 / 31 / 2.568 (p 0.015)
+ K4A-D = SHIP8 (R35)	drop a picture neither listener names, unless DASM clearly hears it	28 / 21 / 2.282	23 / 29 / 2.568 (p 0.017)
+ merge gap 2.5 s	continuing sounds with short breaks no longer drew a second picture	28 / 20 / 2.254	23 / 28 / 2.545
+ grouping ≤ 8 s (R47)	Qwen3-Omni: same continuing sound or a new event?	28 / 18 / 2.197	22 / 26 / 2.545
+ shortest sound 0.3 s (R48)	short real sounds (a laugh burst) were thrown away	29 / 18 / 2.141	22 / 26 / 2.545
+ depict-event (R57)	drop a picture whose event is visibly happening	29 / 17 $\rightarrow$ 18 ^a / 2.141	22 / 26 / 2.545
= B (SHIP8+MD3)	base of the last rounds	29 / 18 / 2.141	22 / 26 / 2.545
+ witness rule + scene check (R53+R60) = D	weak sounds need DASM, both listeners, or one listener and a plausible scene	29 / 14 / 2.028 ^b	24 / 23 / 2.386
+ logit readout (R60L) = D'	scene check read as a probability, no cut-off answers	29 / 15 / 2.056	24 / 24 / 2.409 (p 0.034 vs B)

^a Round 57 scored 29/17 with 26 listener answers missing from a cache; after the cache was filled (1 Oct evening) the same system scores 29/18/2.141, so its net development effect is zero. ^b D's 2.028 contains one gain caused by a cut-off answer ("based on the") that removed a wrong *Glass* picture; the logit readout of D' cannot be cut off, so that picture returns (+1 wrong). The pass bar of Round 60L was set against B, not against D, for this reason (prereg lines 4504–4542).

What the table shows. From the old pipeline to D' the development set gained 11 hits and lost 36 wrong pictures (cost 3.690 $\rightarrow$ 2.056). The test set gained 3 hits and lost 16 wrong pictures

(2.909 $\rightarrow$ 2.409). About a third of the development gain in cost carried over to the test set; with well over a hundred development variants tried, some of the development gain is expected to be selection luck (“winner’s curse”), and the test numbers are the better estimate of what a new video would see.

6.4 What was tried and rejected

Most ideas failed. Table 7 groups the main ones by the question they tried to answer, with the reason from their own pre-registered test. All were decided on the development set or the held-out 415.

Table 7: Main rejected ideas (selection). Hits / wrong / cost on merged DEV unless stated.

idea (round)	what it did	why it failed
<i>Hear more sounds</i>		
other detectors (Sep, rounds 1–8)	FLAM, PretrainedSED (5 backbones), AST, CED, SSLAM, EAT, Dasheng, PE-A-Frame, CLAP verifier, tagger ensembles	recall rose only with many more false spans; PretrainedSED lost three times; ensembles mis-scaled
source separation (Sep, R7)	Demucs, SAM-Audio in front of BEATs	separated audio is out of the detectors’ training distribution: recall under speech fell 9.5 % $\rightarrow$ 4.8 %; SAM-Audio failed its quality check (music never removed)
listener as verifier (R13)	ask Qwen3-Omni yes/no, multiple choice, open list on every weak candidate	finds 5–8 dropped real sounds but adds about 3 wrong pictures per hit
newer listeners (R31–34)	MOSS-Audio, SpotSound, Kimi, Step-Audio	3–10 wrong per extra hit; SpotSound said “Yes” to everything
EXPECT chain (R40–42b)	the scene VLM or a listener names sounds expected off screen; FineLAP times them; DASM confirms	best DEV 31/26/2.254 (+3 hits), but TEST 25/35/2.614 (worse); the gate cannot refuse a wrong off-screen name
short twins, band lists (R49, R56)	add weak or short detections both models half-hear	BANDLIST 0 hits, +6 wrong; TWIN-SHORT 27/36/2.761
tagger ensemble (R63)	calibrated EAT/SSLAM ensemble on BEATs windows	on the 415: 33 % fewer spans but recall 0.152 vs 0.197
<i>Remove wrong pictures</i>		
stricter witnesses (R53 variants)	require DASM or both listeners for every weak sound	27/13/2.113: –5 wrong but 2 real hits lost (fails the 3:1 clause)
named veto, relabel gate (R39–41)	drop or rename a picture when the gate or the listeners name a different sound	real hits lost for no cost gain
held-out family prior (R41)	drop families a detector gets wrong > 70 % of the time on the 415	every affected picture had been heard by a listener: no change
<i>Better on-screen decision</i>		
other gate signals (Sep, am. 14–20)	OWLv2 or SAM 3 vote, video input, synchrony, “any stretch” rule	each removed exactly as many needed sounds as wrong pictures it saved (break-even at $\beta = 2$)
SSL-SaN, SigLIP, BOX (R30–38)	audio-visual localiser; picture-vs-frame similarity; VLM draws a box on the source and checks the crop	each un-silenced real visible sounds; BOX-2 in the full pipeline 29/26/2.366

idea	what it did	why it failed
synchrony (R38)	Synchformer: does on-screen motion move with the sound?	as a veto, +3 needed kept but 5 visible calls lost; as an “add seen” vote, on the bar but changed no shipped picture
human-style chain (HUMAN, R50, R66)	name the source, find it, zoom, ask “is it making the sound now?”	the VLM names “nothing” on most stretches; NAME-ALL passed its gate screen but lost 3 real hits in the pipeline (26/12/2.141)
sign/event and logit gate (R54, R62)	ask for visible effects; read the gate as a logit margin	letter-position bias or “no” habit; best AUROC 0.647, below the 0.65 bar
look-alike reasoning (R61)	“no train here, so it is the washing machine”	works on the laundromat, but also explains away real sounds (a crying sound “explained” by a visible parrot)
bell rule (R64)	concealed actions (a bell in a tower) count as not visible	passed DEV (30/14/1.972) but fires only on bells: not general, not shipped
<i>Timing and grouping</i>		
change-point onsets (R33)	SEBB onset post-processing	7 hits lost
earliest onset (R38), repeat rules (R31, R35)	start at the first detector to hear a sound; require silence before a repeat	hits lost or no net gain
pause questions (R51, R55)	ask whether a 2-s gap is a pause or a new sound	neither the listener nor the detectors can tell a pause from a dropout
context ± 5 s (R59)	give the listener more context	it names what it <i>sees</i> , not what it hears

The main lesson. Two patterns recur in this table. First, the missing sounds can often be found – a listener asked about the whole clip names the hammer, the train, the bell – but every way of adding them brings two to four wrong pictures per hit, because nothing downstream can tell a right off-screen name from a wrong one. Second, every change to the gate moves along the same trade-off line: it silences more on-screen sounds only by silencing about as many needed ones. Section 8 measures both.

6.5 Pictures

The picture model changed four times: Openverse retrieval (v1), SDXL and SDXL-Turbo, FLUX.1-schnell (chosen over PixArt- Σ on twelve real prompts, 13 Sept), and Qwen-Image-2512 (frozen 25 Sept, shipped 28 Sept). On 9 Sept a retrieval-versus-SDXL ablation on the same 100 clips favoured retrieval by +0.17 (gated) and +0.18 (blind) judge points, but 55 % of retrieved images carried a non-commercial or no-derivatives licence and 595 uses drew on only 60 distinct images, so generation was kept. The decisive picture test was a blind *glance test* by the author on 25 Sept: 54 sounds from 50 never-annotated clips, each picture shown for 1.5 s at 384 px with no name, and the author typed what the sound was. FLUX was recognised 14/54 times; Qwen-Image with the same text 26/54 (+0.222 [+0.074, +0.370]); Qwen-Image with a richer subject text 32/54, but that text invented an object (a thud drawn as a door) and was rejected. One of the 26 Qwen-Image pictures was later found to send a false message (a telephone bell drawn as a desk bell), so the +22-point result is reported as “not clean”. A planned sealed confirmation sitting was cancelled on 28 Sept and replaced by the check-and-redraw loop of Section 5.6. A wording test the same day kept the hand-written subject table: 7 of 7 known cases right, against 3 of 7 for two generic VLM wording methods (docs/picture_sense_test_2026-09-28.md).

6.6 Evaluation history: the judge era

The proposal planned an automatic evaluation: a VLM describes the augmented video, an LLM writes a reference of what the viewer should learn, and an independent LLM scores the description against the reference on a 0–4 scale. It was built and run, and it failed in instructive ways:

1. **A reference written by the system’s own models is circular.** The first reference was written by the system’s VLM from the system’s detections; it rewarded whichever system repeated the detector (blind 2.94 > caption 2.68 > gated 2.41 on 9 Sept). A second judge of another family agreed with the first (quadratic $\kappa = 0.753$), which shows only that two judges can share a bias.
2. **A reference written by independent models cannot decide visibility.** A reference built by an audio LLM, CLAP, a different VLM and a different LLM agreed with the human “nothing is missing” label on 55 of 100 clips – chance. An audio-visual model asked with and without the soundtrack reached 49.5 % balanced accuracy (bar 67.7 %); a “gap-closing” score had AUROC 0.529 (bar 0.75).
3. **The rubric was asymmetric.** A withheld picture scored 0 where one was due, a redundant one about 3 where none was due, so the rubric punished staying silent – the very behaviour the gate exists for. A capped rubric (19 Sept) and a human-grounded reference fixed the direction but not the noise.
4. **Judges cannot tell a picture from a text tag.** The best judge (Gemma-4-31B seeing the pictures directly, passing all three trust checks against the author’s labels) scored pictures and text tags alike (-0.04 [$-0.22, +0.14$]), because it reads a tag as the label. Six automatic picture checkers also failed calibration against the author (for example $\kappa = 0.21$ to 0.45 against bars of 0.5).

A five-reviewer scoring panel on 19 Sept moved the primary score to per-sound matching against human labels (Section 3) and kept the judge as a secondary instrument. The answer to RQ4 is built on this history.

7 Results

All numbers in this section are raw counts from the scorer (`benchmark/gold/score_per_sound.py`) or values derived from those counts by Eq. 1–2; derived values are marked. D’ counts are reproduced exactly by the Decision Inspector export (`docs/inspector2/data_parity.json`, `"exact": true`).

7.1 Headline

Table 8: Main result on the merged sets. Recall = hits / needed; precision = hits / (hits + wrong) (derived). “Show nothing” is the cost of a system that never shows a picture, $4 \cdot \text{needed}/\text{clips}$ (derived).

system	merged DEV (71 clips, 58 needed)				merged TEST (88 clips, 65 needed)			
	hits	wrong (v/c/p)	cost	R / P	hits	wrong (v/c/p)	cost	R / P
show nothing	0	0	3.268	–	0	0	2.955	–
old pipeline (B0r)	18	51 (5/37/9)	3.690	.31 / .26	21	40 (2/31/7)	2.909	.32 / .34
B (SHIP8+MD3)	29	18 (6/9/3)	2.141	.50 / .62	22	26 (4/17/5)	2.545	.34 / .46
D (text readout)	29	14 (6/6/2)	2.028	.50 / .67	24	23 (4/14/5)	2.386	.37 / .51
D’ (frozen)	29	15 (6/7/2)	2.056	.50 / .66	24	24 (4/15/5)	2.409	.37 / .50

v / c / p = wrong pictures whose source is on screen / that belong to another sound / with nothing there. Sources: `docs/review/detector_final_2026-10-02.md`; `docs/review/improvements_log_2026-09-30.md` (B0r); `docs/prereg_round13_detector_push.md` lines 3885–3888, 4219–4243, 4536–4544, 4631–4633.

Reading the table.

- On the development set D' finds half of the needed sounds and two out of three of its pictures are right. On the test set it finds 37 % and half of its pictures are right.
- The old pipeline was *worse than showing nothing* on the development set (3.690 vs 3.268) and only slightly better on the test set (2.909 vs 2.955). D' is clearly cheaper than showing nothing on both (2.056 vs 3.268; 2.409 vs 2.955).
- On the test set most wrong pictures of D' belong to another sound (15 of 24); pictures of on-screen sounds are few (4).

7.2 Statistical comparisons

Table 9: Paired clip bootstrap on merged TEST (2000 draws). d = mean per-clip cost difference (negative = first system cheaper); p one-sided as in Section 3.4; the two-sided value is about twice as large.

comparison	d	95 % interval	one-sided p
D' vs B	-0.136	[-0.295, 0.000]	0.034
D vs B	-0.159	[-0.341, -0.023]	0.0145
SHIP8 vs old pipeline B0r	-0.341	[-0.636, -0.045]	0.017
SHIP7 vs B0r	-0.341	[-0.636, -0.045]	0.015
first pre-registered exposure: TO1+F7F8 vs B0r	-0.091	[-0.250, +0.045]	0.132
D' vs B0r	<i>not yet computed</i> (see below)		

Three cautions apply to Table 9. (1) The p -values are one-sided: D' versus B is $p \approx 0.07$ two-sided, so “D' is cheaper than B on new data” is suggestive, not established. (2) The test set had been read after every shipped change (Section 9); none of these comparisons is a blind confirmation. (3) The direct test of D' against the old pipeline has not been run: the per-clip rows of D' on TEST are on the cluster (job 31708734), and the old pipeline's rows were computed with a different display rule (repeat merge 2.0s, no grouping), which moves its wrong-picture count (Section 7.3). The command to run it is in Section 11. The raw difference is large (-0.500 per clip; 3 more hits, 16 fewer wrong pictures), and SHIP8 alone already differs from B0r with one-sided $p = 0.017$, but the number itself is not claimed here.

The step-by-step test reads of the early chain were also corrected for multiplicity (Holm over the four reads of 30 Sept): none of them was significant after correction (`benchmark/gold/stack_stats.json`).

7.3 Sensitivity

On-screen weight. With Eq. 2, D' costs 1.887 / 1.972 / 2.056 on DEV and 2.318 / 2.364 / 2.409 on TEST at $w = 0/1/2$ (derived). For every shipped step from the old pipeline to SHIP8 the cost falls at every w , so no shipping decision depends on how on-screen pictures are weighted (`benchmark/gold/visible_weight_sweep.md`).

Repeat-merge gap. Re-reading the same saved pictures at different merge gaps (`benchmark/gold/merge_gap_sens.py`) moves wrong-picture counts but not the ranking:

gap (s)	SHIP8 DEV	B0r DEV	SHIP8 TEST	B0r TEST
1.5	28 / 23 / 2.338	18 / 57 / 3.859	23 / 30 / 2.591	21 / 44 / 3.000
2.0	28 / 21 / 2.282	18 / 51 / 3.690	23 / 29 / 2.568	21 / 40 / 2.909
3.0	28 / 18 / 2.197	17 / 43 / 3.521	22 / 27 / 2.568	20 / 33 / 2.795

The old pipeline benefits more from a longer gap than SHIP8 does (its many repeated pictures are merged), which is why D' (gap 2.5) against B0r (gap 2.0) is not a like-for-like comparison.

The 0.5 s/1.0 s window and β . The window and β were fixed before the detector push and not varied for D'. In the September table (Section 7.6) the gated system was the cheapest of all systems for any β between 0.39 and 2.56 on TEST (0.46–2.33 on DEV), so $\beta = 2$ lies inside that range.

7.4 Where the misses and wrong pictures come from

The Decision Inspector records, for every needed sound, the step at which it was lost (`docs/review/inspector_trail_ready.md`):

Table 10: Fate of the missed needed sounds of D'.

	DEV	TEST	where
misses	29	41	
lost at a named step	20	32	DEV: band rescue 6, gate 5, DASM vote 2, family merge 2, mirror veto 2, scene check 1, masked-weak 1, one rescue per family 1. TEST: band rescue 6, continuation veto 5, family merge 4, gate 3, mirror veto 3, DASM vote 2, BEATs extraction 2, seven other steps 1 each
never heard by any detector	6	6	no span of the family near the onset
timing / matching	3	3	a same-family span exists but starts outside the window

Two things stand out. Most misses are not “never heard”: the sound was a candidate and a later rule removed it, most often because the listeners did not confirm a weak detection (band rescue). And the gate silences few needed sounds (5 DEV, 3 TEST), but it is also the reason the on-screen wrong pictures remain (6 DEV, 4 TEST).

7.5 How far could this pipeline go? (ceiling analysis)

The ceiling analysis replaces one stage at a time by the gold answer on the *same* detections, using the saved pictures of SHIP7 (`benchmark/gold/ceiling_ship7.md`; this ran before the visibility re-check, so it counts 55 needed sounds and SHIP7 scores 25 / 27 / 2.451 there). It is the best available map for D', which is SHIP7 plus later display and witness rules.

Table 11: Single-stage oracles on SHIP7 (merged DEV, 55 needed).

oracle (one stage made perfect)	hits	wrong (v/c/p)	cost	Δ
none (SHIP7 as shipped)	25	27 (9/16/2)	2.451	–
perfect on-screen gate	30	18 (0/16/2)	1.915	–0.536
perfect listener, filters lifted	34	24 (7/15/2)	1.859	–0.592
perfect vetoes / filters	30	27 (9/16/2)	2.169	–0.282
perfect timing	26	20 (11/7/2)	2.197	–0.254
perfect family	25	26 (14/10/2)	2.423	–0.028
all stages perfect (ceiling)	43	3 (0/1/2)	0.761	–1.690

Applied in sequence (timing, family, gate, vetoes, listener), the gate gives the largest single step (–0.676). Even with every stage perfect, 12 of 55 needed sounds stay missed: 5 are heard by no model at all (a hammer, an explosion, a clang, two golf whacks) and 7 are too faint or too badly timed. This is the *detector floor*: with today’s detectors no decision layer can find more than about four in five needed sounds.

7.6 The value of the on-screen gate (September test table)

The last time the gated system was compared with the *blind* baseline (same detector and generator, no gate) was the September test table (amendment 21, tag `test_final_v33`, TEST-60; `benchmark/gold/holm_test_final_v33_test_bench.json`). It used the old pipeline’s detector, not D’s, but the same gate model and gate rule, so it is the best available estimate of what the gate itself is worth:

Table 12: September test table: gated minus blind on TEST-60, two-sided p with Holm correction.

measure	gated – blind	95 % interval	Holm p
F1 (pre-registered primary)	+0.059	[−0.030, +0.144]	0.18 (unadjusted, null)
precision	+0.137		0.010
wrong pictures per clip	−0.517	[−0.800, −0.283]	< 0.001
viewer cost per clip	−0.833		0.012
clean-clip accuracy	+0.219		0.006
unseen clips: gated vs showing nothing	−3.077		< 0.001
all clips: gated vs showing nothing	−0.233		0.585

The gate removes about half a wrong picture per clip at the price of a small loss in recall, on both halves of the data (on DEV the precision gain was not significant after Holm, 0.080). Its value is therefore real but lies almost entirely in what it does *not* show, which is why F1 – which weighs a miss and a wrong picture equally – does not move. A gate-versus-blind comparison for D’ itself was not re-run.

7.7 Other measurements

Gate accuracy. On the gold DEV sounds the Qwen3.8-27B gate has balanced accuracy 0.617: it silences 16 of 43 on-screen sounds (0.37) and keeps 31 of 36 needed ones (0.86) (`docs/review/vlm_panel_2026-10-01/f1_audit.md`). Its predecessor Qwen2.5-VL-7B reached 0.61. A 0.5-s shift of the sampled frames changes 6 of 79 verdicts (7.6 %).

Listener agreement as evidence. On the held-out 415 AudioSet-Strong clips, a BEATs detection that neither listener names is a real sound 22 % of the time, one named by one listener 37 %, and one named by both 62 %. This is the basis of the witness rule, and it is why the remaining decisions sit where the models disagree at about chance. The audio part of the best rejected idea (a listener names sounds over the whole clip, FineLAP times them, DASM confirms) was right on 74 of 92 detections (0.80) on the same clips, against 0.38–0.39 for the raw detectors: the extra sounds are mostly real, and they become wrong pictures in the full pipeline because the gate passes off-screen ones whose name is wrong, or because the gold lists only the salient sounds.

Onset timing (v3 era). On 255 DCASE 2025 Task 3 events, of the events BEATs detects, hysteresis plus occlusion places the onset within 0.5 s for 80 % (mean absolute error 0.35 s; `benchmark/eval_dcase_onset.json`). Only 26 % of those short-clip events reached the display bar at all, which again points to detection, not timing, as the limit.

Pictures. Section 6.5: in a blind glance test, Qwen-Image pictures were recognised 26 of 54 times against 14 of 54 for FLUX (+0.222 [+0.074, +0.370]), by the author as rater.

8 Error analysis

This section explains *why* the remaining errors happen, using the per-item ledgers (`benchmark/gold/ledger_SHIP8_MD3_WW5.json`), the VLM gate audit of 1 Oct (`docs/review/vlm_panel_2026-10-01/`), the two expert panels of 1 Oct (`docs/review/panel_2026-10-01/`, `docs/review/panel2_2026-10-01/`) and the Decision Inspector.

8.1 Failure modes

1. **Presence taken for production.** The gate’s commonest error: an object being in the frame is taken as that object making the sound now. Examples from the development set: a church tower is visible, so the ringing bell is silenced although nothing in the frames shows it ringing; a golfer stands over the ball, so the whack that happens later is called visible; perched macaws are taken as the source of a bird call made off screen. In the audit this class explains 23 of 36 gate errors across development and test. Two of the three votes effectively measure presence (“name the thing” echoes the label on 30 of 54 development “seen” stretches, and the follow-up “could that thing make it?” is answered from world knowledge), so the majority vote measures presence too.
2. **The “nothing” escape.** The naming question allows “nothing”, and the VLM answers it on 42 % of the stretches the gold calls seen (32 of 76): the maker is small, hidden, or its action is fine (a hand, a coin), the camera is the walker (footsteps), or only the rain is visible during thunder. These sounds keep their picture; this explains 32 of the 38 development gate misses (wrongly kept on-screen sounds).
3. **Masking.** Needed sounds under speech or music are scored low by BEATs, whose top labels there are *Speech* or *Music*. FlexSED hears more of them but mostly below its bar; the listener rescue recovers some, at a price in wrong pictures (Section 6).
4. **Cross pictures.** On the test set 15 of 24 wrong pictures belong to another sound. Examples from the development ledger: a gunshot picture over footsteps, a crowd picture over laughter, a scream picture over a bird call. These are mostly confident detector errors with a plausible scene, which neither the witness rule nor the gate can refuse.
5. **Timing and repeats.** A sound that pauses for more than 2 s and returns is a new labelled event; the system either merges it (a miss) or draws a late second picture of a long sound (a cross). Neither the listener nor the detectors can tell a 2-s pause from a dropout (Rounds 51, 55).
6. **Read-out artefacts.** Asked to answer “(a)” or “(b)”, the 27B model favours a position (“(a)” on 88 % of answers in Round 54); asked yes/no, it has a “no” habit; asked for free text, its answer can be cut off. The fix used in D’ is to read the yes/no logits of the first token, minus those of a negated twin question. This removes the artefacts but not the underlying weakness: as a gate signal the logit margin reaches AUROC 0.647, under the pre-set 0.65 bar.

8.2 The central limit: “present” is not “making the sound”

Every attempt to improve the gate in the project landed on the same trade-off line. In September an OWLv2 vote, a SAM 3 vote and a looser silence rule each removed exactly two wrong pictures for every real needed sound they lost, which is break-even at $\beta = 2$. In the October rounds a box-and-crop check, an audio-visual synchrony model, an audio-visual localiser, a human-style “name, find, zoom, ask” chain and a larger or different VLM each silenced more on-screen sounds only by silencing about as many needed ones. A model swap from Qwen2.5-VL-7B to Qwen3.8-27B moved balanced accuracy from 0.61 to 0.62, and Gemma-4-31B moved along the same line. The VLM audit concluded “not the model”: what is asked of the frames – *is this object making this*

sound right now? – is a question about an action at an instant, and still frames from open VLMs verify presence well and action poorly.

Two observations make the limit concrete. First, the gold’s own notion of *obvious* encodes inference the gate does not do: a bell in a tower is *not* obvious (one cannot see it ring), thunder under visible rain *is*. A rule that treats “concealed actions” as not visible passed the development test (Round 64: 30 / 14 / 1.972) but in practice fired only on bells, and was not shipped as a general rule. Second, a larger vision model (Qwen3-VL-235B) was the only untested lever named by every panel; it does not fit the available disk and is left as future work. The audit’s own reading is that a larger model should be tested on the presence-versus-production question, not assumed to fix it.

8.3 The detector floor

The ceiling analysis (Section 7.5) puts a hard limit on any decision layer: about one needed sound in eleven is heard by no model at all, and about one in eight more is too faint or too mistimed to recover. Every attempt to hear more of them (Section 6.4) found real sounds – 80% of what the best listener chain proposed on the held-out set was real – but brought two to four wrong pictures per new hit on the video benchmark. The reason is the combination of the two limits: a newly heard sound still has to pass the gate, and the gate cannot tell a right off-screen sound from a wrong one.

8.4 Is the gold too strict?

Two human checks bound the effect of the strict “wrong” rule. In Round 46 the author rated, blind, samples of “wrong” pictures: about one in six (2/11 and 3/20) is a real, wanted, off-screen sound the gold does not list. In Round 46b, 12 current wrong pictures were rated: 4 not heard, 5 real but with the source on screen, 2 real and off-screen but timed away from the gold row, 1 unsure; no clean case for a gold edit was found. The ceiling analysis adds that the 14 borderline gate-side items sit one judgement away from the other class: flipping all of them would bring SHIP7’s cost from 2.451 to about 1.9 with no change to the system. The absolute costs are therefore somewhat pessimistic for every system; the comparisons between systems are not affected, because the same gold scores all of them.

8.5 A worked example

Clip `bell_miami` (development) has one needed sound: a *Bell* from 0.2 to 14.5s, importance 3, labelled not visible. Its trail in the Decision Inspector (`docs/inspector2/data.js`) reads as follows. Stage 4 keeps several *Bell* and *Church bell* candidates. The gate cuts the 14-s burst into three stretches of about 4.8s. In each, the name vote answers “church bell” (it shares the word *bell* with the label: yes), the a/b vote says not visibly happening (no), and the description vote says yes. Two of three in every stretch: the sound is silenced, and the needed bell is a miss lost at the gate. The frames show the church; they do not show the bell swinging. This single sound is the whole development gain of the bell rule of Round 64 (“a concealed action is not visible”), and a clean example of presence taken for production: the one vote that asks about the action got it right, and was outvoted by two votes that measure presence.

9 Disclosures and threats to validity

This section collects in one place everything a reader needs to weigh the numbers. Each item says what happened and what it means for the claims.

D1. The test set is a seen set. The merged TEST (88 clips) was read after every shipped change – by count about 19 times between 30 Sept and 2 Oct (the first, pre-registered read; one read per ship SHIP2–SHIP8; reads of five candidates that passed DEV and were not

shipped; the grouping, shortest-sound, depict-event, D and D' reads), plus re-reads of saved test pictures for sensitivity tables. The older TEST-60 part had already been read 11 times before the merge. No choice was made to improve a test number, with two exceptions that were undone (next item); still, choosing among many development variants and reading the test set after each makes the test numbers optimistic. They are a report on a seen set, not a blind confirmation.

- D2. Two test-informed choices, both undone.** A 4-s limit for the grouping rule was set after reading the test result of the unlimited version and was reverted to the pre-registered 8 s; a picture-confidence floor suggested by a test read was not adopted. Rejections of three development passes (EXPECT-A4, EXPECT-A5, AGREE-EARS) cited their test reads alongside held-out and development reasons.
- D3. One-sided p -values.** All detector-push p -values are one-sided bootstrap shares; D' versus B is 0.034 one-sided, about 0.07 two-sided. The direct test of D' against the old pipeline has not been run (Section 7.2).
- D4. Split overlap in September.** A second split file used by the stage-level detector bench (`benchmark/gold/split.json`, DEV-79 / TEST-60) shares 35 of its development clips with the scorer's TEST-60; the FlexSED bar 0.8 was chosen on that DEV-79. On the clean DEV-49 the 0.8 bar passes two of three adoption rules and misses the third by 0.011.
- D5. One annotator, two visibility re-checks.** Every label is the author's; no inter-rater agreement exists. The author re-checked visibility twice on the development set (3 of 14 disputed sounds changed, 0 of 20 random controls changed after a correction); the re-check targeted sounds where the pipeline and the gold disagreed, so it can favour the pipeline. Test labels were never changed.
- D6. Batch 3 tagged with AudioSet labels visible.** Batch 3 of the tagger clips (AudioSet-Strong evaluation clips, 19 clips across DEV2 and TEST2) was labelled with the AudioSet labels available behind a button (shown on 16 of 19), and DASM's checkpoint was selected on that AudioSet split. Earlier batches showed detector suggestions (37 of 50 tagger clips), and benchmark clips were pre-filled from the system under test.
- D7. "Wrong" is strict.** A wrong picture includes redundant pictures of on-screen sounds and late second pictures of a long sound, not only false ones. Round 46 found that about 1 in 6 "wrong" pictures is a real, wanted, off-screen sound missing from the gold. Absolute costs are pessimistic for every system.
- D8. The cost weights are post hoc.** $\beta = 2$ was declared for a gate sweep in September but became the reported metric only after per-sound F1 came out null; it was assumed, not measured with viewers.
- D9. Scored and live configurations differ in four known ways** (kinship direction, picture end cap, subject path, picture model; Section 5.8). The kinship fix in particular was not re-scored for D'.
- D10. Three development sounds are unreachable** by construction (labels the filter never draws), and the ceiling and visible-weight analyses were run on predecessors of D' (SHIP7, SHIP8).
- D11. The picture checker is not calibrated against a person.** Earlier panel rulings said no automatic picture checker should reject pictures until it matched the author's ratings; the shipped check-and-redraw loop was validated on known cases only and uses the same VLM that writes the subject. The planned sealed human confirmation of the pictures was cancelled. The picture claim rests on one blind glance test by the author.
- D12. The author is also the rater** of the glance test and the gold-scope checks. A second annotator packet exists but was not returned.

- D13. No study with deaf or hard-of-hearing viewers was run.** Every claim about benefit to viewers is modelled through the cost weights.
- D14. Larger vision model not tested.** Qwen3-VL-235B, the one lever every panel named for the gate, did not fit the available disk and is future work only.
- D15. Inconsistent records found while writing this report.** The pre-registration log is long (546 KB) and was written under time pressure; the fact-check for this report found small inconsistencies, none of which changes a shipped decision: two records of B’s test row (23/28 and 22/26, both cost 2.545, from before and after the grouping limit was reverted), two records of B’s development wrong split (6/9/3 and 6/10/2, same total), a stale configuration comment (“ ≤ 4 s” for an 8-s value), and the thesis chapter drafts of 27 Sept, which describe an earlier system and are superseded by this report.

10 Limitations and future work

10.1 Limitations

- **Small benchmark.** 71 development and 88 test clips; one or two pictures decide a development comparison, and the test set can detect only cost differences of about 0.19 per clip.
- **One annotator; no viewer study** (Section 9, D5, D13).
- **Detector floor.** About one needed sound in eleven is heard by no model, mostly under speech or music.
- **Gate limit.** Balanced accuracy about 0.62; presence is taken for production; “nothing” is used as an escape.
- **Runtime.** The shipped pipeline is heavy: listener inputs take about 5 minutes per clip on an H200, a picture about 20s per try (up to five tries), and the gate and scene check call a 27B model many times. It is an offline tool, not a live captioning system.
- **Vocabulary.** The 527 AudioSet classes and the 215 drawable families have no word for some sounds (“phone alert”); the scorer’s single-parent ontology file ignores 38 multi-parent labels.
- **Licences.** Most clips are for research use and are not redistributed; Audio Flamingo Next has a non-commercial licence.

10.2 Future work

In order of expected value:

1. **A study with DHH viewers**, as designed in docs/beta_specification.md: 5–10 viewers, one question per clip (“helped / hurt / neither”), and an ordinal regression of the answer on misses and wrong pictures, which measures β instead of assuming it.
2. **A second annotator** for 30 clips (the packet exists), with κ on the needed decision; below 0.4 the definition itself would need work.
3. **The presence-versus-production question:** test a larger VLM (Qwen3-VL-235B) on exactly this question; a production veto read as a logit margin on the named phrase; a temporal contrast (the onset frame against a control window); an external grounding box with a crop.
4. **A detector that hears under speech and music**, trained on mixtures, tested on the unread fresh set of 422 AudioSet-Strong clips.

5. **Benchmark v2:** at least 300 clips, at least two annotators, the two tagger-guide rules (2-s return, visibility split) written into the guideline, and a sealed test split with a submission scorer.
6. **Display:** a picture with a word, and a “tentative” style for low-confidence pictures, both of which need viewers to evaluate.

11 Reproduction

11.1 Code state

Check out the tag `detector-frozen-2026-10-02`. `config.use_shipped()` sets `D'`; `config.use_scored()` sets the old pipeline `B0r`; the scoring harness names every variant in `ARMS` of `benchmark/gold/round13_dev.py` (lineage: `T01+F7F8` line 152 $\rightarrow$... $\rightarrow$ `SHIP8+MD3+WW5+SL` line 243). The cluster copies used for scoring are not git repositories; the frozen code was synced from the tag.

11.2 Environment and models

- Conda environment `msproj`: PyTorch 2.5.1, transformers 5.16.1 (Qwen3.x VLMs need ≥ 5.8), diffusers, faster-whisper, pannels-inference, easyocr, sentence-transformers. FineLAP needs transformers 4.51.3 and runs in a separate `~/venv_flap` called as a subprocess. The ComfyUI environment is recorded in `docs/envs/comfy.yml`. The `requirements.txt` file predates the final stack and is incomplete.
- FlexSED and DASM run from their own repositories (`~/FlexSED`, `Transformer4SED`), imported in process.
- Model identifiers (Hugging Face): `Qwen/Qwen3.8-27B`; `Qwen/Qwen3-Omni-30B-A3B-Instruct`; `nvidia/audio-flamingo-next-hf`; `AndreasXi/FineLAP` (revision `b419aa2`); `CPF2/detect_any_sound`; `Qwen/Qwen-Image-2512`; `google/owlv2-base-patch16-ensemble`; BEATs iter3+AS2M cpt2; PANNs CNN14. Revisions other than FineLAP are not pinned; the Hugging Face cache was used offline.
- Hardware: Bar-Ilan SLURM cluster, partitions with H200 and A100 GPUs. Qwen-Image needs about 57 GB in bf16; the picture check needs the generator and the VLM on one H200 or two cards.

11.3 Running one video

```
python main.py --input clip.mp4 --device cuda
```

`main.py` applies `use_shipped()`. Without `-listener-split`, `src/listener_prep.py` first builds the per-clip inputs the detector needs (FlexSED scores, a first pass with the old pipeline, BEATs and PANNs scores, the listener pools and answers, DASM and FineLAP scores) – about 5 minutes on an H200 – and then runs `pipeline.run`. The output is `data/output/<stem>_augmented.mp4`.

11.4 Running a folder of clips on the cluster

```
sbatch slurm/run_best.sh NAME /path/to/clips      # stages 1-5
python scripts/best_to_protocol.py --name NAME --arm SHIP8+MD3+WW5+SL
python scripts/repaint_shipped.py --tag NAMEbest_v33 --verify
python scripts/recomposite.py --root data/work/shipped_v_NAMEbest_v33 \
    --shipped --require-local-images --modes clean --out OUT
```

The batch runner stops at the list of pictures; the three scripts draw and compose them.

11.5 Reproducing the scores

```
python benchmark/gold/round13_dev.py stage4    # then stage5, score
python benchmark/gold/merged_dev.py
python benchmark/gold/test_vs_ship8.py
python benchmark/gold/parity_check.py SHIP8+MD3+WW5+SL
python benchmark/gold/inspector_trail_export.py \
    --expect DEV=29/58/15/6/7/2/2.056 --expect TEST=24/65/24/4/15/5/2.409
```

The last command re-runs D' with logging on every decision and fails unless the counts match exactly; it passed on 2 Oct (0 of 159 clips differ). To compute the missing D'-versus-old-pipeline test, run `benchmark/gold/final_test.py` (score) with the D' arm on the cluster, where both per-clip rows exist.

11.6 The Decision Inspector

`docs/inspector2/` is a static page over `data.js` (12.8 MB, 159 clips) that shows, for every gold sound and every candidate of D', each step it passed or failed, the value, the bar and the model answers, next to the rendered video (`docs/inspector/media/bysig/`, 350 MB, not in git). The logging hooks (`src/trail.py`) only record and never change a decision. Known gaps: 86 of 156 gate decisions reuse stored verdicts without the raw answers; the depict-event look-alike replies are not stored.

11.7 Where the records are

file	content
<code>docs/prereg_round13_detector_push.md</code>	every pre-registration and result, rounds 13–66
<code>docs/prereg_v4.md</code> , <code>docs/prereg_round*.md</code>	v4 amendments 1–25, rounds 4–12
<code>docs/review/detector_final_2026-10-02.md</code>	plain-language summary of D' and the last day
<code>docs/review/improvements_log_2026-09-30.md</code>	ship chain and rejected ideas, 30 Sept – 1 Oct
<code>docs/review/panel*_2026-10-01/</code> , <code>docs/panel_*.md</code>	expert panels
<code>benchmark/gold/annotations/gold_AG.json</code>	the gold labels
<code>benchmark/gold/ledger_*.json</code>	per-item misses and wrong pictures
<code>benchmark/gold/ceiling_ship7.md</code> , <code>visible_weight_sweep.md</code>	ceiling and sensitivity analyses
<code>docs/LEDGER_2026-09-26.md</code>	audit ledger of the September phase

12 Conclusion

A deaf viewer of a video misses the sounds whose sources are not on screen, and no existing channel supplies them automatically. This project built a pipeline that does – detection by two detectors checked by two listeners and a witness model, an on-screen decision from the frames, a guarded picture beside the video – from open models with no training; a per-sound benchmark of 159 scored clips; and a metric that prices what a viewer loses. The frozen system finds half of the needed sounds on the development set and 37 % on the test set, with two thirds and one half of its pictures right, and is far cheaper than both showing nothing and the September pipeline. The remaining errors sit where today's open models reach their limit: sounds no model hears under speech and music, and a vision model that takes “the object is there” for “the object is making the sound”.

Answers to the research questions.

RQ1 – which audio information matters most?

Discrete, sourced events whose source is off screen: sirens, alarms, bells, a baby crying, glass, a knock. The benchmark’s importance rule (an event one can say in a sentence, or danger) captures them; steady place noise (wind, traffic hum) is excluded by design, and it is also where detectors, gate and labels disagree most. Whether DHH viewers rank sounds the same way is untested.

RQ2 – what granularity?

Decide and score at the level of the AudioSet *family* (depth ≥ 1), draw the specific source. Family matching was the one de-duplication and scoring rule that generalised without a threshold; the picture names the maker (a barking dog, not “bark”) from a hand table, which beat generic VLM wording 7/7 to 3/7.

RQ3 – can generated augmentations improve accessibility beyond subtitles?

Not shown. What is shown is that the system adds information no subtitle carries (off-screen sounds) at a modelled cost below showing nothing, and that its pictures are recognised at a glance about half the time. A study with viewers is the necessary next step.

RQ4 – how to evaluate?

Per sound, against human labels, with the error split by cause and a cost that prices a miss against a wrong picture, reported over a range of weights; decisions pre-registered on a development set. Automatic protocols with model-written references or model judges failed: they are circular, decide visibility at chance, or cannot tell a picture from a text tag.

RQ5 – comparison with audio-to-visual approaches?

Audio-to-image models draw the whole scene and cannot leave a visible object out, so they were represented by a blind baseline with the same detector and generator. Against it, the gate removes about half a wrong picture per clip at a small cost in recall (Holm-significant on the September test table); F1 does not move because the gate’s value is in what it does not show.

A Prompts

All questions are asked with greedy decoding and, for Qwen3.8-27B, thinking off. $\langle \text{label} \rangle$ is the family name, lower-cased and cut before a comma or bracket. Source: `src/stage5_cross_modal_analysis/reason.py`, `benchmark/listener_round.py`, `benchmark/gold/listener_variants.py`, `src/stage6_visual_augmentation/group.py`, `depict.py`, `verify.py`, `gen_screen.py`.

Gate: name vote

“These frames are from the moment a sound of $\langle \text{label} \rangle$ was heard. Name the thing in these frames that is making that sound. Answer with a short noun phrase of at most 5 words. If nothing that could make that sound is visible in these frames, answer exactly: nothing.”

Gate: follow-up (text only)

“Sound heard: $\langle \text{label} \rangle$ / Thing visible in the video: $\langle \text{named} \rangle$ / Could that thing be what is making that sound? Answer yes or no.”

Gate: a/b vote (asked in both orders)

“These frames are from the moment a sound of $\langle \text{label} \rangle$ was heard. Judge from the frames alone. (a) you can SEE $\langle \text{label} \rangle$ happening on screen – the source is in the frames and visibly making that sound (b) $\langle \text{label} \rangle$ is not visibly happening in these frames. Answer with the letter only.”

Gate: describe vote

“Describe what is happening in these frames in one sentence. Name what you see and what it is doing. No more than 25 words.”

Speech context (text only, both orders)

“... Are they (a) reacting to that sound or talking about it, or (b) not referring to that sound?”

Scene check (witness rule; logit read)

“Could the sound of $\langle \text{label} \rangle$ plausibly be heard in this scene? Answer yes or no.” and the twin
 “Could the sound of $\langle \text{label} \rangle$ NOT plausibly be heard in this scene? Answer yes or no.”

Listener yes/no (Qwen3-Omni, AF Next)

“Is the sound of $\langle \text{family} \rangle$ present in this recording? Answer yes or no.”

Listener inventory

“List every distinct non-speech sound you hear in this recording, one per line, most prominent first.” (64 new tokens)

Grouping (Qwen3-Omni, both orders)

“You hear a $\langle \text{lab} \rangle$ sound near the start and again near the end of this recording. Is the $\langle \text{lab} \rangle$ near the end $\langle o_1 \rangle$, or $\langle o_2 \rangle$? Answer with exactly one word: $\langle w_1 \rangle$ or $\langle w_2 \rangle$.” with the options
 “the same continuing sound as at the start (for example one alarm or engine with a short pause)” and “a new, separate event (for example a second bark, knock or shot)”.

Depict-event

“Is “ $\langle \text{event} \rangle$ ” visibly happening in these frames? Answer yes or no.” and “Do these frames show no “ $\langle \text{event} \rangle$ ” happening? ...”; then (text only) “Could the sound of $\langle B \rangle$ be mistaken for the sound of $\langle A \rangle$?” in both directions.

Picture check

“Look at this picture. What is the main thing shown in it? Choose exactly one option. ... Answer with the letter only.”

Picture prompt tail

“, the whole thing fully in frame, caught at the moment it makes the sound with its visible effect, one large subject filling the picture, plain white background”; negative prompt: scenery, landscape, background scene, room interior, street, buildings, sky background, text, letters, watermark.

B Configuration of D'

Values set by `config.use_shipped()` (`config.py`, lines 503–572), read by running it. “ $\rightarrow$ ” gives the module default and the shipped value.

variable	value	meaning
AED_THRESHOLD	0.175	BEATs extraction bar
AED_MIN_DUR	0.5 $\rightarrow$ 0.3	shortest span (s)
DISPLAY_THRESHOLD	0.35	display bar (stage 5)
FLEXSED_BAR	0.8	FlexSED extraction bar
FLEXSED_VETO	0 $\rightarrow$ 0.3	FlexSED clip veto
PANNS_VETO	0.05	PANNS clip veto on FlexSED-only spans
TWIN_MAX	True	twin confidence = stronger side
MIRROR_VETO	0.7	mirror veto bar
LISTENER_CONFIRMED_MIRROR	True	listener can undo the mirror veto
KEEP_NEEDS_V4	True	a keep must also be named in an inventory
MASKED_WEAK_VETO, _AF	True	masked-weak veto
DASM_CLIP_VETO	0.0839	DASM clip veto
DASM_LOCAL_VETO	0.35	witness rule bar

variable	value	meaning
DASM_LOCAL_KEEP	both	both listeners can witness
SCENE_FIT_LOGIT	True	scene check by logit margin
KEEP_NEEDS_V4_ALL	onto	inventory keep incl. descendants
KEEP_NEEDS_V4_ALL_DASM_KEEP	True	DASM can keep it
BAND_TWIN_PULL	0.5	band-twin pull
CONTINUATION_VETO	0.5	continuation veto
LISTENER_RESCUE	True	band rescue
LISTENER_LO	0.4 → 0.5	lowest FlexSED peak rescued
LISTENER_RULE	TIER	Qwen alone ≥ 0.6 , both below
DASM_RESCUE, _NEW_ONLY	True	DASM rescue of new families
FINE LAP_VETO	0.329	FineLAP veto on rescued spans
LISTENER_DASM_VOTE / _BAR	True / 0.575	DASM vote on rescued spans
LISTENER_ONCE	True	one rescue per family per clip
ONSET_CAM, ONSET_MONOTONE	True	occlusion onsets, never earlier
MAX_SPAN	8.0 → None	no length cap
LABEL_FILTER	depictable	what may be drawn
VLM_MODEL	Qwen/Qwen3.8-27B	gate, subject, checks
VLM_THINKING	False	thinking off
VISIBILITY_STRETCH / _RULE	5.0 / majority	gate stretches and vote
KINSHIP_DIRECTED	True (scored: False)	kinship direction
VIDEO_BACKEND	owlv2	stage-2 objects
DEPICT_EVENT	True	depict-event drop
MERGE_GAP	2.0 → 2.5	repeat merge (s)
MIN_DWELL	1.5	minimum on screen (s)
MAX_AFTER_END	1.0 (scored: None)	cap past the real end (s)
GROUP_ASK / GROUP_MAX_GAP	True / 8.0	grouping by Qwen3-Omni
MAX_SLOTS	3	simultaneous pictures
GEN_MODEL	Qwen/Qwen-Image-2512	picture model (FLUX retired)
RESOLUTION	1024 × 1024	picture size
PICTURE_V3, _SCENE, _FINAL, _MAKER	True	subject writing
PICTURE_VERIFY / _TRIES	True / 5	check and redraw
CONFIDENCE_FADE	False	full opacity

C Round-by-round record

One row per round or main arm, in order. Results are hits / wrong / cost on merged DEV (55 or 58 needed, see Section 4) unless marked; “415” = held-out AudioSet-Strong screen; “old DEV” = DEV-49. Full text, bars and line numbers: `docs/prereg_v4.md` (amendments), `docs/prereg_round4..12*.md`, `docs/prereg_round13_detector_push.md` (rounds 13–66). Counted from that file: rounds 13–28 ran 129 new full-pipeline variants (106 on DEV-49 alone), rounds 29–41 ran 21 more and 42 component screens, rounds 42–57 ran 5 full variants and about 30 screens, rounds 58–66 four full variants and 15 screens.

round	idea	result	verdict
<i>September amendments (prereg_v4)</i>			
am. 5	per-sound gold, F1 headline	$\Delta F1 +0.028$ [−0.022, +0.074]; precision, false alarms significant	F1 null

round	idea	result	verdict
am. 8	FlexSED union, bar 0.8	passes 3/3 rules	shipped
am. 9	viewer cost, $\beta = 2$ (post hoc)	gated 3.36 vs blind 4.24; silence 2.79	reported metric
am. 10	FlexSED veto 0.3	4.12 $\rightarrow$ 3.59	shipped
am. 11–15	per-family bars; speech-removed view; family silence; any-stretch rule; OWLv2 vote	each worse or neutral	rejected
am. 16	PANNs veto 0.05	3.02 / 3.10	shipped
am. 17	tagger ensemble	CED mis-scaled	rejected
am. 18–20	visibility signals (video, SAM 3, synchrony)	all at break-even 2:1	rejected
am. 21	final September TEST table	Table 12	reported
am. 22–25	cascade, PE-A-Frame, Qwen3-Omni verifier	17 hits, cost 4.82; AUROC 0.53; AUROC 0.661	rejected
<i>Rounds 4–12 (AudioSet harness, 28 Sept)</i>			
4	BEATs self-veto replaces PANNs veto	415 neutral; later worse on DEV and TEST (2.83 vs 2.63, p 0.043)	reverted 30 Sept
5	EAT, Dasheng	415 +0.087	fail
6	DASM as detector	415 +0.294 [+0.111, +0.511]	fail (kept as witness)
7, 7b	SAM-Audio separation	quality check failed	stop
8	17 ideas (I1–I10)	closest 17 -0.039 , p 0.061	fail
9	local contrast J2	415 pass, DEV 14 $\rightarrow$ 13 hits	not shipped
10	DASM rescue R0–R7	9 of 61 targets reachable	hold
11	masker tests	all worse	no pick
12	cost v2 on AudioSet	shipped 1.650 vs nothing 0.986	harness retired
<i>Rounds 13–28 (29–30 Sept)</i>			
13	listener rescue, many question forms; R13-1 twin-max	~ 3 wrong per extra hit; R13-1 TEST = B0r	R13-1 kept inside TO1
14	TO1+F7F8	old DEV 18/24/2.45 vs 2.78; merged 23/48/3.155; TEST 22/38/2.818 (p 0.13)	shipped (SHIP1)
16	N2b masked-weak veto	23/44/3.042; TEST 22/35/2.750	shipped
19–20	DR2 DASM rescue	24/44/2.986	shipped (SHIP2)
21	K-V4 inventory keep	24/41/2.901	shipped (SHIP3)
22–27	N2c, stacks, K2 variants	N2c 22/38/2.930 lost a hit	rejected
28	DV DASM clip veto	24/34/2.704; TEST 23/34/2.682	shipped (SHIP4)
<i>Rounds 29–41 (30 Sept – 1 Oct)</i>			
29	F8-U, DV-L, DV-G; V4D; GA	+2 hits for +7 wrong; no-op; -5 hits; STOP; STOP	rejected
30	BTP band-twin pull	25/33/2.620; TEST 23/33/2.659	shipped (SHIP5)
30	K4AO; PIC-SIM; RT	held; nothing flips; no-op	held / rejected
31	CONT continuation veto	25/28/2.479; TEST 24/32/2.591	shipped (SHIP6)
31	FineLAP veto 0.329	25/27/2.451; TEST 24/31/2.568	shipped (SHIP7)
31	RPT-S, PMC, N2c, DVG, PTC, SUBJ, OM, SPOT, H2	screens or arms without gain	rejected
32	FineLAP in DASM’s seat; DV2	+11 wrong; Thunder lost	rejected
33	SPOT-V, change-point onsets, K4A-4E, SSL-SaN gate	all “Yes”; -7 hits; no-op; STOP	rejected
34	MOSS-Audio as listener	+4/ -9 needed, +63 other	rejected
35	DBX, DBR	premise wrong; cost not lower	rejected
re-check	visibility re-check (gold)	3/14 flips; control 0/20	gold change
35	K4A-D (K4AO + DASM keep)	28/58, 21/2.282; TEST 23/29/2.568 (p 0.017)	shipped (SHIP8)
36–37	DASM impact peaks + VLM naming; BOX gate	+2 cross; -7 seen silenced	rejected
38	CF, BOX-2, GBTP, SHIP8+BOX2, SYNC, SYNC-2	best BOX-2 arm 29/26/2.366	rejected
39	DETACHED-ADD, CONTRAST, MAKER-VIS, RELABEL-GATE	+7 wrong; +142 other; -9 hits; hit lost	rejected
40–40e	EXPECT chain	A4: DEV 31/26/2.254, TEST 25/35/2.614	not shipped
41	AGREE, NAMED-VETO, PRIOR415	below bar; hits lost; no change	rejected
<i>Rounds 42–57 (1 Oct)</i>			
42	held-out check of EXPECT-A4 audio chain	74/92 = 0.80 precise vs 0.38–0.39 raw	diagnostic
42b	EXPECT-A5 (7 vouched families)	DEV 30/21/2.169; TEST 23/31/2.614	not shipped

round	idea	result	verdict
43–43c, 50	gate gold-set variants (HUMAN, HUMAN-2, HUMAN-ADD)	one sound short of the bar at best	rejected
44	AVNAME (eyes + ears name the sound)	rename 15/34; drop 27/17/2.225, TEST no change	rejected
45	AGREE-EARS (second listener family)	DEV 31/24/2.197; 415 0.833 < 0.85; TEST 2.659	rejected
46, 46b	gold scope (blind rating of wrong pictures)	≈ 1 in 6 is a real unlisted sound	no gold edit
–	merge gap 2.5 s	28/20/2.254	shipped
47	GRP-A grouping by Qwen3-Omni, ≤ 8 s	28/18/2.197	shipped
48	MD3 shortest span 0.3 s	29/18/2.141	shipped (= B)
49	BANDLIST	0 hits, +6 wrong	rejected
51, 52b, 55	pause rules, kin-keep, frame-pause	415 STOP	rejected
52	SK7 strong keep	29/22/2.254	rejected
53, 53d	WEAK-WITNESS alone	27/13/2.113 (2 hits lost); 27/17/2.225	fail alone
53b, 53c	witness bars from the 415	outside allowed range	stop
54, 54b	SIGN (visible effect)	0 flips; “no” on 9/10	rejected
56	TWIN-SHORT	27/36/2.761	rejected
57	DEPICT-EVENT	29/17 (net 0 after cache fill)	shipped
<i>Rounds 58–66 (1–2 Oct)</i>			
58, 58b	tight listener cut; FlexSED witness	415 precision 0.400, 0.297	stop
59	context ± 5 s	415 0.333	stop
60	SCENE-MARGIN + witness (= D)	29/14/2.028; TEST 24/23/2.386 (p 0.0145)	shipped
61b–d	look-alike reasoning (Qwen, Gemma)	hits lost in every variant	fail
62, 62b	logit gate; PRIOR	AUROC 0.647 / 0.644 < 0.65	stop
60L	scene check by logit margin (= D')	29/15/2.056; TEST 24/24/2.409 (p 0.034)	shipped, frozen
63	tagger ensemble	415 recall 0.152 vs 0.197	stop
64	concealed action {Bell}	30/14/1.972 (one clip)	not general, not shipped
65	perceptual-onset return	415 0.115	stop
66	NAME-ALL crop margin	gate screen passed; DEV 26/12/2.141 (3 hits lost)	fail

D Glossary of internal names

name	meaning
B0, B0r	the old pipeline, <code>config.use_scored()</code> (September test table); B0r = B0 re-run in the same job as a candidate
B1	B0 with the BEATs self-veto in place of the PANNs veto (rejected)
B	SHIP8+MD3: the base of the last rounds (DEV 29/18/2.141)
D, D'	B + witness rule + scene check; D' reads the scene check by logits (frozen)
DEV-49, TEST-60	per-sound gold halves of 22 Sept
DEV2, TEST2	tagger clips added 29 Sept – 1 Oct
merged DEV / TEST	DEV-49 + DEV2 (71 clips); TEST-60 + TEST2 (88 clips)
needed	a sound neither visible nor obvious, importance 2–3
hit, miss	a needed sound with / without a matching picture
wrong: visible / cross / phantom	a picture of an on-screen sound / another sound / nothing
cost	$(4 \cdot \text{missed} + 2 \cdot \text{wrong}) / \text{clips}$
listener	an audio-language model asked about a candidate sound (Qwen3-Omni, Audio Flamingo Next)
V4 / inventory	the listener’s open list of heard sounds

name	meaning
TIER	rescue rule: Qwen alone for strong FlexSED runs, both listeners for weak ones
TO1+F7F8	first shipped listener stack (tiered rescue, once per family, twin-max, mirror veto + listener keep, DASM vote)
N2b	masked-weak veto
DR2	DASM rescue of new families
K-V4	a listener keep must also be named in an inventory
DV	DASM clip veto
BTP	band-twin pull (start a late picture at an earlier weak run)
CONT	continuation veto
FLAP	FineLAP veto on rescued spans
K4AO, K4A-D	drop a picture neither inventory names (K4A-D: unless DASM hears it)
SHIP1-SHIP8	successive shipped configurations (Table 6)
MD3	shortest span $0.5 \rightarrow 0.3$ s
GRP / GROUP	grouping of same-family pictures by Qwen3-Omni
WW, WW5	weak-witness rule; WW5 = WW with the scene check
SL	scene check read as a logit margin (Round 60L)
DEPICT	depict-event drop (Round 57)
415	held-out AudioSet-Strong screening set
EXPECT	idea family: name sounds expected off screen, then confirm (rejected)

References

- [1] O. Alonzo, H. V. Shin, D. Li. Beyond subtitles: Captioning and visualizing non-speech sounds to improve accessibility of user-generated videos. *ASSETS*, 2022. doi:10.1145/3517428.3544808.
- [2] K. Alhamoud et al. Vision-language models do not understand negation (NegBench). *CVPR*, 2025. arXiv:2501.09425.
- [3] Qwen Team (S. Bai et al.). Qwen2.5-VL technical report. arXiv:2502.13923, 2025.
- [4] Black Forest Labs. FLUX.1 [schnell]. Model release, 2024. <https://huggingface.co/black-forest-labs/FLUX.1-schnell>.
- [5] Ç. Bilen, G. Ferroni, F. Tuveri, J. Azcarreta, S. Krstulović. A framework for the robust evaluation of sound event detection. *ICASSP*, 2020. arXiv:1910.08440.
- [6] B. C. Biner, F. M. Sofian, U. B. Karakaş, D. Ceylan, E. Erdem, A. Erdem. SonicDiffusion: Audio-driven image generation and editing with pretrained diffusion models. arXiv:2405.00878, 2024.
- [7] H. Chen, W. Xie, A. Vedaldi, A. Zisserman. VGGSound: A large-scale audio-visual dataset. *ICASSP*, 2020. arXiv:2004.14368.
- [8] H. Chen, W. Xie, T. Afouras, A. Nagrani, A. Vedaldi, A. Zisserman. Localizing visual sounds the hard way. *CVPR*, 2021. arXiv:2104.02691.
- [9] S. Chen, Y. Wu, C. Wang, S. Liu, D. Tompkins, Z. Chen, F. Wei. BEATs: Audio pre-training with acoustic tokenizers. *ICML*, 2023. arXiv:2212.09058.
- [10] Y. Chu, J. Xu, X. Zhou, Q. Yang, S. Zhang, Z. Yan, C. Zhou, J. Zhou. Qwen-Audio: Advancing universal audio understanding via unified large-scale audio-language models. arXiv:2311.07919, 2023.
- [11] Y. Chu et al. Qwen2-Audio technical report. arXiv:2407.10759, 2024.
- [12] Described and Captioned Media Program. Captioning Key: Sound effects and music. <https://dcmp.org/learn/captioningkey>, retrieved 27 Sept 2026.

- [13] C. de Lacerda Patata, S. Hassan, N. Tinker, R. L. Peiris, M. Huenerfauth. Caption Royale: Exploring the design space of affective captions from the perspective of deaf and hard-of-hearing individuals. *CHI*, 2024. doi:10.1145/3613904.3642258.
- [14] C. Drummond, R. C. Holte. Cost curves: An improved method for visualizing classifier performance. *Machine Learning* 65(1):95–130, 2006.
- [15] B. Elizalde, S. Deshmukh, M. Al Ismail, H. Wang. CLAP: Learning audio concepts from natural language supervision. *ICASSP*, 2023.
- [16] G. Ferroni, N. Turpault, J. Azcarreta, F. Tuveri, R. Serizel, Ç. Bilen, S. Krstulović. Improving sound event detection metrics: Insights from DCASE 2020. *ICASSP*, 2021. arXiv:2010.13648.
- [17] D. Ge, S. Liu, C. Gong, X.-L. Zhang, C. Zhang, X. Li. Towards expressive and faithful audio-to-image generation: A unified multimodal dataset and synthesis framework (AudioCanvas). *ACM MM*, 2026. arXiv:2608.09529.
- [18] J. F. Gemmeke, D. P. W. Ellis, D. Freedman, A. Jansen, W. Lawrence, R. C. Moore, M. Plakal, M. Ritter. Audio Set: An ontology and human-labeled dataset for audio events. *ICASSP*, 2017.
- [19] T. Geng, T. Wang, J. Duan, R. Cong, F. Zheng. Dense-localizing audio-visual events in untrimmed videos: A large-scale benchmark and baseline. *CVPR*, 2023. arXiv:2303.12930.
- [20] Y. Gong, Y.-A. Chung, J. Glass. AST: Audio spectrogram transformer. *Interspeech*, 2021. arXiv:2104.01778.
- [21] J. Hai, H. Wang, W. Guo, M. Elhilali. FlexSED: Towards open-vocabulary sound event detection. *WASPAA*, 2025. arXiv:2509.18606.
- [22] S. Hershey, D. P. W. Ellis, E. Fonseca, A. Jansen, C. Liu, R. C. Moore, M. Plakal. The benefit of temporally-strong labels in audio event classification. *ICASSP*, 2021. arXiv:2105.07031.
- [23] D. Jain, K. Mack, A. Amrous, M. Wright, S. Goodman, L. Findlater, J. E. Froehlich. HomeSound: An iterative field deployment of an in-home sound awareness system for deaf or hard of hearing users. *CHI*, 2020. doi:10.1145/3313831.3376758.
- [24] D. Jain, L. Findlater, J. Gilkeson, B. Holland, R. Duraiswami, D. Zotkin, C. Vogler, J. E. Froehlich. Head-mounted display visualizations to support sound awareness for the deaf and hard of hearing. *CHI*, 2015.
- [25] D. Jain, A. Lin, R. Guttman, M. Amalachandran, A. Zeng, L. Findlater, J. Froehlich. Exploring sound awareness in the home for people who are deaf or hard of hearing. *CHI*, 2019.
- [26] X. Juanola, G. Haro, M. Fuentes. A critical assessment of visual sound source localization models including negative audio. *ICASSP*, 2025. arXiv:2410.01020.
- [27] X. Juanola, G. Morais, M. Fuentes, G. Haro. Learning from silence and noise for visual sound source localization. *BMVC*, 2025. arXiv:2508.21761.
- [28] Q. Kong, Y. Cao, T. Iqbal, Y. Wang, W. Wang, M. D. Plumbley. PANNs: Large-scale pretrained audio neural networks for audio pattern recognition. *IEEE/ACM TASLP* 28:2880–2894, 2020.
- [29] H. Liu, C. Li, Q. Wu, Y. J. Lee. Visual instruction tuning. *NeurIPS*, 2023. arXiv:2304.08485.
- [30] T. Matthews, J. Fong, F. W.-L. Ho-Ching, J. Mankoff. Evaluating non-speech sound visualizations for the deaf. *Behaviour & IT* 25(4):333–351, 2006.
- [31] L. May, K. Ohshiro, K. Dang, S. Sridhar, J. Pai, M. Fuentes, S. Lee, M. Cartwright. Unspoken sound: Identifying trends in non-speech audio captioning on YouTube. *CHI*, 2024. doi:10.1145/3613904.3642162.
- [32] L. May, M. Clemens, K. Dang, K. Ohshiro, S. Sridhar, P. Wee, M. Fuentes, S. Lee, M. Cartwright. “Choices? That’s the dream”: Challenges and opportunities in non-speech information closed-captioning. *Frontiers in Computer Science* 7, 2025.

- [33] A. Mesaros, T. Heittola, T. Virtanen. Metrics for polyphonic sound event detection. *Applied Sciences* 6(6):162, 2016.
- [34] A. Owens, A. A. Efros. Audio-visual scene analysis with self-supervised multisensory features. *ECCV*, 2018. arXiv:1804.03641.
- [35] D. Petermann, M. M. Kalayeh. Seeing Sound: Assembling sounds from visuals for audio-to-image generation. arXiv:2501.05413, 2025.
- [36] D. Podell et al. SDXL: Improving latent diffusion models for high-resolution image synthesis. arXiv:2307.01952, 2023.
- [37] C. de Lacerda Pataca, M. Watkins, R. Peiris, S. Lee, M. Huenerfauth. Visualization of speech prosody and emotion in captions: Accessibility for deaf and hard-of-hearing users. *CHI*, 2023. doi:10.1145/3544548.3581511.
- [38] D. Jain et al. ProtoSound: A personalized and scalable sound recognition system for deaf and hard-of-hearing users. *CHI*, 2022. arXiv:2202.11134.
- [39] P. Revuelta, T. Ortiz, M. J. Lucía, B. Ruiz, J. M. Sánchez-Pena. Limitations of standard accessible captioning of sounds and music for deaf and hard of hearing people: An EEG study. *Frontiers in Integrative Neuroscience* 14, 2020.
- [40] S. Sakshi et al. MMAU: A massive multi-task audio understanding and reasoning benchmark. arXiv:2410.19168, 2024.
- [41] A. Senocak, T.-H. Oh, J. Kim, M.-H. Yang, I. S. Kweon. Learning to localize sound source in visual scenes. *CVPR*, 2018.
- [42] A. Senocak, H. Ryu, J. Kim, T.-H. Oh, H. Pfister, J. S. Chung. Sound source localization is all about cross-modal alignment. *ICCV*, 2023. arXiv:2309.10724.
- [43] D. Jain, H. Ngo, P. Patel, S. Goodman, L. Findlater, J. Froehlich. SoundWatch: Exploring smartwatch-based deep learning approaches to support sound awareness for deaf and hard of hearing users. *ASSETS*, 2020.
- [44] K. Sung-Bin, A. Senocak, H. Ha, A. Owens, T.-H. Oh. Sound to visual scene generation by audio-to-visual latent alignment. *CVPR*, 2023. arXiv:2303.17490.
- [45] C. Tang et al. SALMONN: Towards generic hearing abilities for large language models. *ICLR*, 2024. arXiv:2310.13289.
- [46] E. Tzinis, S. Wisdom, A. Jansen, S. Hershey, T. Remez, D. P. W. Ellis, J. R. Hershey. Into the wild with AudioScope: Unsupervised audio-visual separation of on-screen sounds. *ICLR*, 2021. arXiv:2011.01143.
- [47] S. J. Um, D. Kim, S. Lee, J. U. Kim. Object-aware sound source localization via audio-visual scene understanding. *CVPR*, 2025. arXiv:2506.18557.
- [48] A. Vyas et al. Pushing the frontier of audiovisual perception with large-scale multimodal correspondence learning. arXiv:2512.19687, 2025.
- [49] S. Wiedenbeck. The use of icons and labels in an end user application program: An empirical study of learning and retention. *Behaviour & IT* 18(2):68–82, 1999.
- [50] T.-H. Wu, J. E. Gonzalez, T. Darrell, D. M. Chan. CLAIR-A: Leveraging large language models to judge audio captions. *ASRU*, 2025. arXiv:2409.12962.
- [51] Y. Wu et al. FLAM: Frame-wise language-audio modeling. *ICML*, 2025. arXiv:2505.05335.
- [52] C. Wu et al. Qwen-Image technical report. arXiv:2508.02324, 2025.
- [53] J. Xu et al. Qwen3-Omni technical report. arXiv:2509.17765, 2025.
- [54] G. Yariv, I. Gat, L. Wolf, Y. Adi, I. Schwartz. AudioToken: Adaptation of text-conditioned diffusion models for audio-to-image generation. *Interspeech*, 2023. arXiv:2305.13050.

- [55] G. Yariv, I. Gat, S. Benaim, L. Wolf, I. Schwartz, Y. Adi. Diverse and aligned audio-to-video generation via text-to-video model adaptation. *AAAI*, 2024. arXiv:2309.16429.
- [56] M. Yoshida, R. Togo, T. Ogawa, M. Haseyama. Off-screen sound separation based on audio-visual pre-training using binaural audio. *Sensors* 23(9):4540, 2023.
- [57] L. Zheng et al. Judging LLM-as-a-judge with MT-Bench and Chatbot Arena. *NeurIPS Datasets and Benchmarks*, 2023. arXiv:2306.05685.
- [58] C. Zheng, H. Zhou, F. Meng, J. Zhou, M. Huang. Large language models are not robust multiple choice selectors. *ICLR*, 2024. arXiv:2309.03882.
- [59] Z. Zhou, Z. Zhang, X. Xu, Z. Xie, M. Wu, K. Q. Zhu. Can audio captions be evaluated with image caption metrics? *ICASSP*, 2022.